



? In flash photography, the energy used to make the flash is stored in a capacitor, which consists of two closely spaced conductors that carry opposite charges. If the amount of charge on the conductors is doubled, by what factor does the stored energy increase? (i) $\sqrt{2}$; (ii) 2; (iii) $2\sqrt{2}$; (iv) 4; (v) 8.

24 Capacitance and Dielectrics

When you stretch the rubber band of a slingshot or pull back the string of an archer's bow, you are storing mechanical energy as elastic potential energy. A capacitor is a device that stores *electric* potential energy and electric charge. To make a capacitor, just insulate two conductors from each other. To store energy in this device, transfer charge from one conductor to the other so that one has a negative charge and the other has an equal amount of positive charge. Work must be done to move the charges through the resulting potential difference between the conductors, and the work done is stored as electric potential energy.

Capacitors have a tremendous number of practical applications in devices such as electronic flash units for photography, mobile phones, airbag sensors for cars, and radio and television receivers. We'll encounter many of these applications in later chapters (particularly Chapter 31, in which we'll see the crucial role played by capacitors in the alternating-current circuits that pervade our technological society). In this chapter, however, our emphasis is on the fundamental properties of capacitors. For a particular capacitor, the ratio of the charge on each conductor to the potential difference between the conductors is a constant, called the *capacitance*. The capacitance depends on the sizes and shapes of the conductors and on the insulating material (if any) between them. Compared to the case in which there is only vacuum between the conductors, the capacitance increases when an insulating material (a *dielectric*) is present. This happens because a redistribution of charge, called *polarization*, takes place within the insulating material. Studying polarization will give us added insight into the electrical properties of matter.

Capacitors also give us a new way to think about electric potential energy. The energy stored in a charged capacitor is related to the electric field in the space between the conductors. We'll see that electric potential energy can be regarded as being stored *in the field itself*. The idea that the electric field is itself a storehouse of energy is at the heart of the theory of electromagnetic waves and our modern understanding of the nature of light, to be discussed in Chapter 32.

LEARNING OUTCOMES

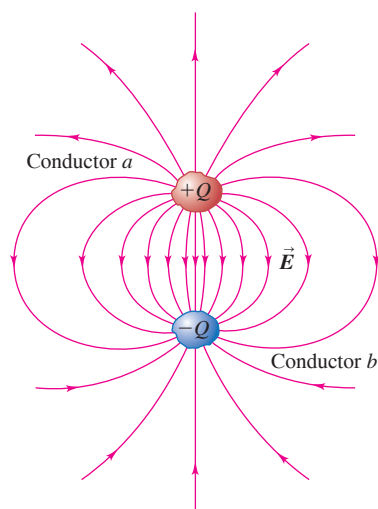
In this chapter, you'll learn...

- 24.1** The nature of capacitors, and how to calculate a quantity that measures their ability to store charge.
- 24.2** How to analyze capacitors connected in a network.
- 24.3** How to calculate the amount of energy stored in a capacitor.
- 24.4** What dielectrics are, and how they make capacitors more effective.
- 24.5** How a dielectric inside a charged capacitor becomes polarized.
- 24.6** How to use Gauss's laws when dielectrics are present.

You'll need to review...

- 21.2, 21.5, 21.7** Polarization; field of charged conductors; electric dipoles.
- 22.3–22.5** Gauss's law.
- 23.3, 23.4** Potential for charged conductors; potential due to a cylindrical charge distribution.

Figure 24.1 Any two conductors a and b insulated from each other form a capacitor.



24.1 CAPACITORS AND CAPACITANCE

Any two conductors separated by an insulator (or a vacuum) form a **capacitor** (Fig. 24.1). In most practical applications, each conductor initially has zero net charge and electrons are transferred from one conductor to the other; this is called *charging* the capacitor. Then the two conductors have charges with equal magnitude and opposite sign, and the *net* charge on the capacitor as a whole remains zero. We'll assume throughout this chapter that this is the case. When we say that a capacitor has charge Q , or that a charge Q is *stored* on the capacitor, we mean that the conductor at higher potential has charge $+Q$ and the conductor at lower potential has charge $-Q$ (assuming that Q is positive). Keep this in mind in the following discussion and examples.

In circuit diagrams a capacitor is represented by either of these symbols:



The vertical lines (straight or curved) represent the conductors, and the horizontal lines represent wires connected to either conductor. One common way to charge a capacitor is to connect these two wires to opposite terminals of a battery. Once the charges Q and $-Q$ are established on the conductors, the battery is disconnected. This gives a fixed *potential difference* V_{ab} between the conductors (that is, the potential of the positively charged conductor a with respect to the negatively charged conductor b) that is just equal to the voltage of the battery.

The electric field at any point in the region between the conductors is proportional to the magnitude Q of charge on each conductor. It follows that the potential difference V_{ab} between the conductors is also proportional to Q . If we double the magnitude of charge on each conductor, the charge density at each point doubles, the electric field at each point doubles, and the potential difference between conductors doubles; however, the *ratio* of charge to potential difference does not change. This ratio is called the **capacitance** C of the capacitor:

$$\text{Capacitance of a capacitor } C = \frac{Q}{V_{ab}} \quad (24.1)$$

$\leftarrow$ Magnitude of charge on each conductor
 $\leftarrow$ Potential difference between conductors (a has charge $+Q$, b has charge $-Q$)

The SI unit of capacitance is called one **farad** (1 F), in honor of the 19th-century English physicist Michael Faraday. From Eq. (24.1), one farad is equal to one *coulomb per volt* (1 C/V):

$$1 \text{ F} = 1 \text{ farad} = 1 \text{ C/V} = 1 \text{ coulomb/volt}$$

CAUTION Capacitance vs. coulombs Don't confuse the symbol C for capacitance (which is always in italics) with the abbreviation C for coulombs (which is never italicized).

The greater the capacitance C of a capacitor, the greater the magnitude Q of charge on either conductor for a given potential difference V_{ab} and hence the greater the amount of stored energy. (Remember that potential is potential energy per unit charge.) Thus *capacitance is a measure of the ability of a capacitor to store energy*. We'll see that the capacitance depends only on the shapes, sizes, and relative positions of the conductors and on the nature of the insulator between them. (For special types of insulating materials, the capacitance *does* depend on Q and V_{ab} . We won't discuss these materials, however.)

Calculating Capacitance: Capacitors in Vacuum

We can calculate the capacitance C of a given capacitor by finding the potential difference V_{ab} between the conductors for a given magnitude of charge Q and then using Eq. (24.1). For now we'll consider only *capacitors in vacuum*; that is, empty space separates the conductors that make up the capacitor.

The simplest form of capacitor consists of two parallel conducting plates, each with area A , separated by a distance d that is small in comparison with their dimensions (Fig. 24.2a). When the plates are charged, the electric field is almost completely localized in the region between the plates (Fig. 24.2b). As we discussed in Example 22.8 (Section 22.4), the field between such plates is essentially *uniform*, and the charges on the plates are uniformly distributed over their opposing surfaces. We call this arrangement a **parallel-plate capacitor**.

We found the electric-field magnitude E for this arrangement in Example 21.12 (Section 21.5) by using the principle of superposition of electric fields and again in Example 22.8 (Section 22.4) by using Gauss's law. It would be a good idea to review those examples. We found that $E = \sigma/\epsilon_0$, where σ is the magnitude (absolute value) of the surface charge density on each plate. This is equal to the magnitude of the total charge Q on each plate divided by the area A of the plate, or $\sigma = Q/A$, so the field magnitude E can be expressed as

$$E = \frac{\sigma}{\epsilon_0} = \frac{Q}{\epsilon_0 A}$$

The field is uniform and the distance between the plates is d , so the potential difference (voltage) between the two plates is

$$V_{ab} = Ed = \frac{1}{\epsilon_0} \frac{Qd}{A}$$

Thus

$$\text{Capacitance of a parallel-plate capacitor in vacuum} \quad C = \frac{Q}{V_{ab}} = \epsilon_0 \frac{A}{d} \quad (24.2)$$

Magnitude of charge on each plate
Area of each plate
Distance between plates
Electric constant

Potential difference between plates

The capacitance depends on only the geometry of the capacitor; it is directly proportional to the area A of each plate and inversely proportional to their separation d . The quantities A and d are constants for a given capacitor, and ϵ_0 is a universal constant. Thus in vacuum the capacitance C is a constant independent of the charge on the capacitor or the potential difference between the plates. If one of the capacitor plates is flexible, C changes as the plate separation d changes. This is the operating principle of a condenser microphone (Fig. 24.3).

When matter is present between the plates, its properties affect the capacitance. We'll return to this topic in Section 24.4. Meanwhile, we remark that if the space contains air at atmospheric pressure instead of vacuum, the capacitance differs from the prediction of Eq. (24.2) by less than 0.06%.

In Eq. (24.2), if A is in square meters and d in meters, then C is in farads. The units of the electric constant ϵ_0 are $\text{C}^2/\text{N} \cdot \text{m}^2$, so

$$1 \text{ F} = 1 \text{ C}^2/\text{N} \cdot \text{m} = 1 \text{ C}^2/\text{J}$$

Because $1 \text{ V} = 1 \text{ J/C}$ (energy per unit charge), this equivalence agrees with our definition $1 \text{ F} = 1 \text{ C/V}$. Finally, we can express the units of ϵ_0 as $1 \text{ C}^2/\text{N} \cdot \text{m}^2 = 1 \text{ F/m}$, so

$$\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$$

This relationship is useful in capacitance calculations, and it also helps us to verify that Eq. (24.2) is dimensionally consistent.

Figure 24.2 A charged parallel-plate capacitor.

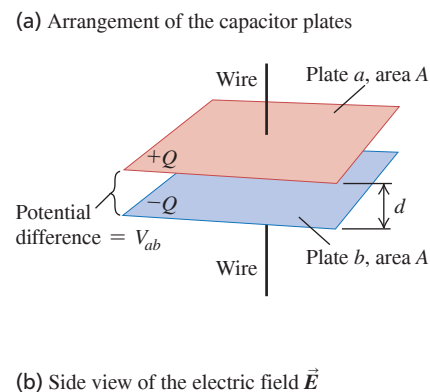
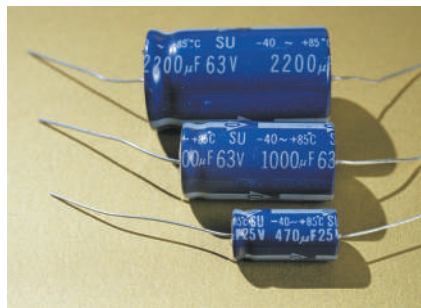


Figure 24.3 Inside a condenser microphone is a capacitor with one rigid plate and one flexible plate. The two plates are kept at a constant potential difference V_{ab} . Sound waves cause the flexible plate to move back and forth, varying the capacitance C and causing charge to flow to and from the capacitor in accordance with the relationship $C = Q/V_{ab}$. Thus a sound wave is converted to a charge flow that can be amplified and recorded digitally.



Figure 24.4 A commercial capacitor is labeled with the value of its capacitance. For these capacitors, $C = 2200\ \mu\text{F}$, $1000\ \mu\text{F}$, and $470\ \mu\text{F}$.



One farad is a very large capacitance, as the following example shows. In many applications the most convenient units of capacitance are the *microfarad* ($1\ \mu\text{F} = 10^{-6}\ \text{F}$) and the *picofarad* ($1\ \text{pF} = 10^{-12}\ \text{F}$). For example, the flash unit in a point-and-shoot camera uses a capacitor of a few hundred microfarads (Fig. 24.4), while capacitances in a radio tuning circuit are typically from 10 to 100 picofarads.

For *any* capacitor in vacuum, the capacitance C depends only on the shapes, dimensions, and separation of the conductors that make up the capacitor. If the conductor shapes are more complex than those of the parallel-plate capacitor, the expression for capacitance is more complicated than in Eq. (24.2). In the following examples we show how to calculate C for two other conductor geometries.

CAUTION **Capacitor charge and potential difference** Saying that a capacitor carries charge Q means that one conductor has positive charge $+Q$ and the other conductor has negative charge $-Q$. The *net* charge on the capacitor is zero! Furthermore, the positively charged conductor is always at higher potential than the negatively charged conductor. If the potential difference V_{ab} between the conductors is 500 V, the potential of the positive conductor is 500 V greater than that of the negative conductor. !

EXAMPLE 24.1 Size of a 1 F capacitor

The parallel plates of a 1.0 F capacitor are 1.0 mm apart. What is their area?

IDENTIFY and SET UP This problem uses the relationship among the capacitance C , plate separation d , and plate area A (our target variable) for a parallel-plate capacitor. We solve Eq. (24.2) for A .

EXECUTE From Eq. (24.2),

$$A = \frac{Cd}{\epsilon_0} = \frac{(1.0\ \text{F})(1.0 \times 10^{-3}\ \text{m})}{8.85 \times 10^{-12}\ \text{F/m}} = 1.1 \times 10^8\ \text{m}^2$$

EVALUATE This corresponds to a square about 10 km (about 6 miles) on a side! The volume of such a capacitor would be at least

$Ad = 1.1 \times 10^5\ \text{m}^3$, equivalent to that of a cube about 50 m on a side. In fact, it's possible to make 1 F capacitors a few *centimeters* on a side. The trick is to have an appropriate substance between the plates rather than a vacuum, so that (among other things) the plate separation d can be greatly reduced. We'll explore this further in Section 24.4.

KEYCONCEPT The capacitance of a capacitor is a measure of how much charge its plates will hold for a given potential difference (voltage) between the plates. Capacitance is proportional to the area of the plates and inversely proportional to the distance between them.

EXAMPLE 24.2 Properties of a parallel-plate capacitor

WITH VARIATION PROBLEMS

The plates of a parallel-plate capacitor in vacuum are 5.00 mm apart and $2.00\ \text{m}^2$ in area. A 10.0 kV potential difference is applied across the capacitor. Compute (a) the capacitance; (b) the charge on each plate; and (c) the magnitude of the electric field between the plates.

IDENTIFY and SET UP We are given the plate area A , the plate spacing d , and the potential difference $V_{ab} = 1.00 \times 10^4\ \text{V}$ for this parallel-plate capacitor. Our target variables are the capacitance C , the charge Q on each plate, and the electric-field magnitude E . We use Eq. (24.2) to calculate C and then use Eq. (24.1) and V_{ab} to find Q . We use $E = Q/\epsilon_0 A$ to find E .

EXECUTE (a) From Eq. (24.2),

$$\begin{aligned} C &= \epsilon_0 \frac{A}{d} = (8.85 \times 10^{-12}\ \text{F/m}) \frac{(2.00\ \text{m}^2)}{5.00 \times 10^{-3}\ \text{m}} \\ &= 3.54 \times 10^{-9}\ \text{F} = 0.00354\ \mu\text{F} \end{aligned}$$

(b) The charge on the capacitor is

$$\begin{aligned} Q &= CV_{ab} = (3.54 \times 10^{-9}\ \text{C/V})(1.00 \times 10^4\ \text{V}) \\ &= 3.54 \times 10^{-5}\ \text{C} = 35.4\ \mu\text{C} \end{aligned}$$

The plate at higher potential has charge $+35.4\ \mu\text{C}$, and the other plate has charge $-35.4\ \mu\text{C}$.

(c) The electric-field magnitude is

$$\begin{aligned} E &= \frac{\sigma}{\epsilon_0} = \frac{Q}{\epsilon_0 A} = \frac{3.54 \times 10^{-5}\ \text{C}}{(8.85 \times 10^{-12}\ \text{C}^2/\text{N} \cdot \text{m}^2)(2.00\ \text{m}^2)} \\ &= 2.00 \times 10^6\ \text{N/C} \end{aligned}$$

EVALUATE We can also find E by recalling that the electric field is equal in magnitude to the potential gradient [Eq. (23.22)]. The field between the plates is uniform, so

$$E = \frac{V_{ab}}{d} = \frac{1.00 \times 10^4\ \text{V}}{5.00 \times 10^{-3}\ \text{m}} = 2.00 \times 10^6\ \text{V/m}$$

(Remember that $1\ \text{N/C} = 1\ \text{V/m}$.)

KEYCONCEPT The electric field between the plates of a parallel-plate capacitor is uniform. To find its magnitude E , divide the potential difference between the plates by their separation; E is also proportional to the ratio of the charge on the capacitor plates to the plate's area.

EXAMPLE 24.3 A spherical capacitor

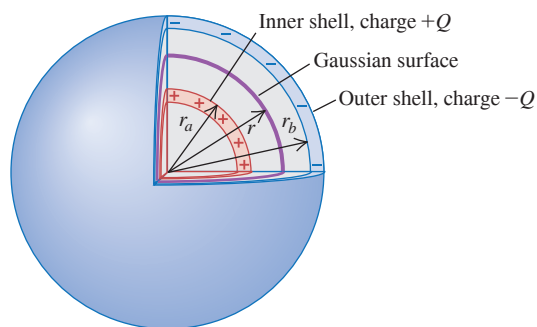
WITH VARIATION PROBLEMS

Two concentric spherical conducting shells are separated by vacuum (Fig. 24.5). The inner shell has total charge $+Q$ and outer radius r_a , and the outer shell has charge $-Q$ and inner radius r_b . Find the capacitance of this spherical capacitor.

IDENTIFY and SET UP By definition, the capacitance C is the magnitude Q of the charge on either sphere divided by the potential difference V_{ab} between the spheres. We first find V_{ab} , and then use Eq. (24.1) to find the capacitance $C = Q/V_{ab}$.

EXECUTE Using a Gaussian surface such as that shown in Fig. 24.5, we found in Example 22.5 (Section 22.4) that the charge on a conducting sphere produces zero field *inside* the sphere, so the outer sphere makes no contribution to the field between the spheres. Therefore the electric field *and* the electric potential between the shells are the same as those outside a charged conducting sphere with charge $+Q$. We considered that problem in Example 23.8 (Section 23.3), so the same result applies here: The potential at any point between the spheres is $V = Q/4\pi\epsilon_0 r$.

Figure 24.5 A spherical capacitor.



Hence the potential of the inner (positive) conductor at $r = r_a$ with respect to that of the outer (negative) conductor at $r = r_b$ is

$$\begin{aligned} V_{ab} &= V_a - V_b = \frac{Q}{4\pi\epsilon_0 r_a} - \frac{Q}{4\pi\epsilon_0 r_b} \\ &= \frac{Q}{4\pi\epsilon_0} \left(\frac{1}{r_a} - \frac{1}{r_b} \right) = \frac{Q}{4\pi\epsilon_0} \frac{r_b - r_a}{r_a r_b} \end{aligned}$$

The capacitance is then

$$C = \frac{Q}{V_{ab}} = 4\pi\epsilon_0 \frac{r_a r_b}{r_b - r_a}$$

As an example, if $r_a = 9.5$ cm and $r_b = 10.5$ cm,

$$\begin{aligned} C &= 4\pi(8.85 \times 10^{-12} \text{ F/m}) \frac{(0.095 \text{ m})(0.105 \text{ m})}{0.010 \text{ m}} \\ &= 1.1 \times 10^{-10} \text{ F} = 110 \text{ pF} \end{aligned}$$

EVALUATE We can relate our expression for C to that for a parallel-plate capacitor. The quantity $4\pi r_a r_b$ is intermediate between the areas $4\pi r_a^2$ and $4\pi r_b^2$ of the two spheres; in fact, it's the *geometric mean* of these two areas, which we can denote by A_{gm} . The distance between spheres is $d = r_b - r_a$, so we can write $C = 4\pi\epsilon_0 r_a r_b / (r_b - r_a) = \epsilon_0 A_{\text{gm}} / d$. This has the same form as for parallel plates: $C = \epsilon_0 A / d$. If the distance between spheres is very small in comparison to their radii, their capacitance is the same as that of parallel plates with the same area and spacing.

KEYCONCEPT To find the capacitance of any capacitor made up of two conductors, first calculate the potential difference V_{ab} between the two conductors when they carry charges $+Q$ and $-Q$. The capacitance is equal to Q divided by V_{ab} .

EXAMPLE 24.4 A cylindrical capacitor

WITH VARIATION PROBLEMS

Two long, coaxial cylindrical conductors are separated by vacuum (Fig. 24.6). The inner cylinder has outer radius r_a and linear charge density $+\lambda$. The outer cylinder has inner radius r_b and linear charge density $-\lambda$. Find the capacitance per unit length for this capacitor.

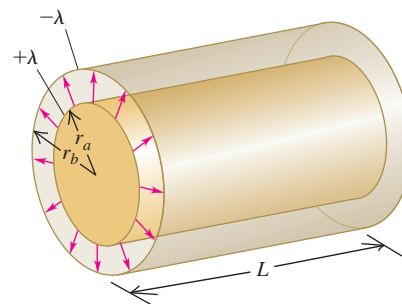
IDENTIFY and SET UP As in Example 24.3, we use the definition of capacitance, $C = Q/V_{ab}$. We use the result of Example 23.10 (Section 23.3) to find the potential difference V_{ab} between the cylinders, and find the charge Q on a length L of the cylinders from the linear charge density. We then find the corresponding capacitance C from Eq. (24.1). Our target variable is this capacitance divided by L .

EXECUTE As in Example 24.3, the potential V between the cylinders is not affected by the presence of the charged outer cylinder. Hence our result in Example 23.10 for the potential outside a charged conducting cylinder also holds in this example for potential in the space between the cylinders:

$$V = \frac{\lambda}{2\pi\epsilon_0} \ln \frac{r_0}{r}$$

Here r_0 is the arbitrary, *finite* radius at which $V = 0$. We take $r_0 = r_b$, the radius of the inner surface of the outer cylinder. Then the potential at the outer surface of the inner cylinder (at which $r = r_a$) is just the potential V_{ab} of the inner (positive) cylinder a with respect to the outer (negative) cylinder b :

Figure 24.6 A long cylindrical capacitor. The linear charge density λ is assumed to be positive in this figure. The magnitude of charge in a length L of either cylinder is λL .



$$V_{ab} = \frac{\lambda}{2\pi\epsilon_0} \ln \frac{r_b}{r_a}$$

If λ is positive as in Fig. 24.6, then V_{ab} is positive as well: The inner cylinder is at higher potential than the outer.

Continued

The total charge Q in a length L is $Q = \lambda L$, so from Eq. (24.1) the capacitance C of a length L is

$$C = \frac{Q}{V_{ab}} = \frac{\lambda L}{\frac{\lambda}{2\pi\epsilon_0} \ln \frac{r_b}{r_a}} = \frac{2\pi\epsilon_0 L}{\ln(r_b/r_a)}$$

The capacitance per unit length is

$$\frac{C}{L} = \frac{2\pi\epsilon_0}{\ln(r_b/r_a)}$$

Substituting $\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m} = 8.85 \text{ pF/m}$, we get

$$\frac{C}{L} = \frac{55.6 \text{ pF/m}}{\ln(r_b/r_a)}$$

EVALUATE The capacitance of coaxial cylinders is determined entirely by their dimensions, just as for parallel-plate and spherical capacitors. Ordinary coaxial cables are made like this but with an insulating material instead of vacuum between the conductors. A typical cable used for connecting a television to a cable TV feed has a capacitance per unit length of 69 pF/m.

KEYCONCEPT The capacitance per unit length of two long, coaxial, conducting cylinders depends on the ratio of the inner radius of the outer conductor to the outer radius of the inner conductor.

Figure 24.7 An assortment of commercially available capacitors.



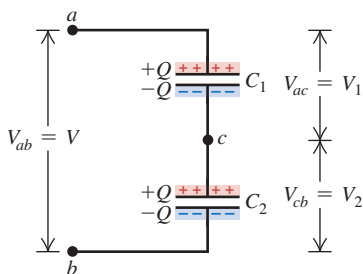
Figure 24.8 A series connection of two capacitors.

(a) Two capacitors in series

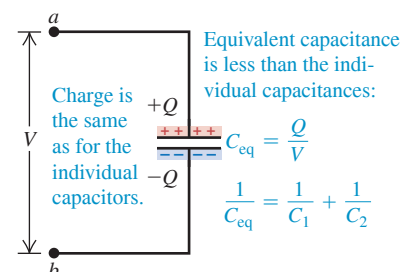
Capacitors in series:

- The capacitors have the same charge Q .
- Their potential differences add:

$$V_{ac} + V_{cb} = V_{ab}$$



(b) The equivalent single capacitor



TEST YOUR UNDERSTANDING OF SECTION 24.1 A capacitor has vacuum in the space between the conductors. If you double the amount of charge on each conductor, what happens to the capacitance? (i) It increases; (ii) it decreases; (iii) it remains the same; (iv) the answer depends on the size or shape of the conductors.

ANSWER

(iii) The capacitance does not depend on the value of the charge Q . Doubling Q causes the potential difference V_{ab} to double, so the capacitance $C = Q/V_{ab}$ remains the same. These statements are true no matter what the geometry of the capacitor.

24.2 CAPACITORS IN SERIES AND PARALLEL

Capacitors are manufactured with certain standard capacitances and working voltages (Fig. 24.7). However, these standard values may not be the ones you actually need in a particular application. You can obtain the values you need by combining capacitors; many combinations are possible, but the simplest combinations are a series connection and a parallel connection.

Capacitors in Series

Figure 24.8a is a schematic diagram of a **series connection**. Two capacitors are connected in series (one after the other) by conducting wires between points a and b . Both capacitors are initially uncharged. When a constant positive potential difference V_{ab} is applied between points a and b , the capacitors become charged; the figure shows that the charge on *all* conducting plates has the same magnitude. To see why, note first that the top plate of C_1 acquires a positive charge Q . The electric field of this positive charge pulls negative charge up to the bottom plate of C_1 until all of the field lines that begin on the top plate end on the bottom plate. This requires that the bottom plate have charge $-Q$. These negative charges had to come from the top plate of C_2 , which becomes positively charged with charge $+Q$. This positive charge then pulls negative charge $-Q$ from the connection at point b onto the bottom plate of C_2 . The total charge on the lower plate of C_1 and the upper plate of C_2 together must always be zero because these plates aren't connected to anything except each other. Thus *in a series connection the magnitude of charge on all plates is the same*.

Referring to Fig. 24.8a, we can write the potential differences between points a and c , c and b , and a and b as

$$V_{ac} = V_1 = \frac{Q}{C_1}, \quad V_{cb} = V_2 = \frac{Q}{C_2}, \quad V_{ab} = V = V_1 + V_2 = Q \left(\frac{1}{C_1} + \frac{1}{C_2} \right)$$

and so

$$\frac{V}{Q} = \frac{1}{C_1} + \frac{1}{C_2} \quad (24.3)$$

Following a common convention, we use the symbols V_1 , V_2 , and V to denote the potential differences V_{ac} (across the first capacitor), V_{cb} (across the second capacitor), and V_{ab} (across the entire combination of capacitors), respectively.

The **equivalent capacitance** C_{eq} of the series combination is defined as the capacitance of a *single* capacitor for which the charge Q is the same as for the combination, when the potential difference V is the same. In other words, the combination can be replaced by an *equivalent capacitor* of capacitance C_{eq} . For such a capacitor, shown in Fig. 24.8b,

$$C_{eq} = \frac{Q}{V} \quad \text{or} \quad \frac{1}{C_{eq}} = \frac{V}{Q} \quad (24.4)$$

Combining Eqs. (24.3) and (24.4), we find

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2}$$

We can extend this analysis to any number of capacitors in series. We find the following result for the *reciprocal* of the equivalent capacitance:

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \cdots \quad (24.5)$$

Capacitors in series: Equivalent capacitance of series combination Capacitances of individual capacitors

The reciprocal of the equivalent capacitance of a series combination equals the sum of the reciprocals of the individual capacitances. In a series connection the equivalent capacitance is always *less than* any individual capacitance.

CAUTION Capacitors in series The magnitude of charge is the same on all plates of all the capacitors in a series combination; however, the potential differences of the individual capacitors are *not* the same unless their individual capacitances are the same. The potential differences of the individual capacitors add to give the total potential difference across the series combination: $V_{\text{total}} = V_1 + V_2 + V_3 + \cdots$

Capacitors in Parallel

The arrangement shown in **Fig. 24.9a** is called a **parallel connection**. Two capacitors are connected in parallel between points a and b . In this case the upper plates of the two capacitors are connected by conducting wires to form an equipotential surface, and the lower plates form another. Hence *in a parallel connection the potential difference for all individual capacitors is the same* and is equal to $V_{ab} = V$. The charges Q_1 and Q_2 are not necessarily equal, however, since charges can reach each capacitor independently from the source (such as a battery) of the voltage V_{ab} . The charges are

$$Q_1 = C_1 V \quad \text{and} \quad Q_2 = C_2 V$$

The *total* charge Q of the combination, and thus the total charge on the equivalent capacitor, is

$$Q = Q_1 + Q_2 = (C_1 + C_2)V$$

so

$$\frac{Q}{V} = C_1 + C_2 \quad (24.6)$$

The parallel combination is equivalent to a single capacitor with the same total charge $Q = Q_1 + Q_2$ and potential difference V as the combination (Fig. 24.9b). The equivalent capacitance of the combination, C_{eq} , is the same as the capacitance Q/V of this single equivalent capacitor. So from Eq. (24.6),

$$C_{eq} = C_1 + C_2$$

APPLICATION Touch Screens and Capacitance

The touch screen on a mobile phone, a tablet, or (as shown here) a medical device uses the physics of capacitors. Behind the screen are two parallel layers, one behind the other, of thin strips of a transparent conductor such as indium tin oxide. A voltage is maintained between the two layers. The strips in one layer are oriented perpendicular to those in the other layer; the points where two strips overlap act as a grid of capacitors. When you bring your finger (a conductor) up to a point on the screen, your finger and the front conducting layer act like a second capacitor in series at that point. The circuitry attached to the conducting layers detects the location of the capacitance change, and so detects where you touched the screen.

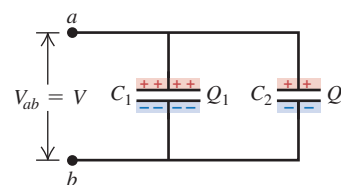


Figure 24.9 A parallel connection of two capacitors.

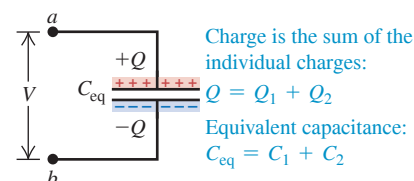
(a) Two capacitors in parallel

Capacitors in parallel:

- The capacitors have the same potential V .
- The charge on each capacitor depends on its capacitance: $Q_1 = C_1 V$, $Q_2 = C_2 V$.



(b) The equivalent single capacitor



In the same way we can show that for any number of capacitors in parallel,

$$\text{Capacitors in parallel: } C_{\text{eq}} = C_1 + C_2 + C_3 + \cdots \quad (24.7)$$

Equivalent capacitance of parallel combination Capacitances of individual capacitors

The equivalent capacitance of a parallel combination equals the *sum* of the individual capacitances. In a parallel connection the equivalent capacitance is always *greater than* any individual capacitance.

CAUTION Capacitors in parallel The potential differences are the same for all capacitors in a parallel combination; however, the charges on individual capacitors are *not* the same unless their individual capacitances are the same. The charges on the individual capacitors add to give the total charge on the parallel combination: $Q_{\text{total}} = Q_1 + Q_2 + Q_3 + \cdots$. [Compare these statements to those in the “Caution” paragraph following Eq. (24.5).]

PROBLEM-SOLVING STRATEGY 24.1 Equivalent Capacitance

IDENTIFY *the relevant concepts:* The concept of equivalent capacitance is useful whenever two or more capacitors are connected.

SET UP *the problem* using the following steps:

1. Make a drawing of the capacitor arrangement.
2. Identify all groups of capacitors that are connected in series or in parallel.
3. Keep in mind that when we say a capacitor “has charge Q ,” we mean that the plate at higher potential has charge $+Q$ and the other plate has charge $-Q$.

EXECUTE *the solution* as follows:

1. Use Eq. (24.5) to find the equivalent capacitance of capacitors connected in series, as in Fig. 24.8. Such capacitors each have the *same charge* if they were uncharged before they were connected; that charge is the same as that on the equivalent capacitor. The potential difference across the combination is the sum of the potential differences across the individual capacitors.

2. Use Eq. (24.7) to find the equivalent capacitance of capacitors connected in parallel, as in Fig. 24.9. Such capacitors all have the *same potential difference* across them; that potential difference is the same as that across the equivalent capacitor. The total charge on the combination is the sum of the charges on the individual capacitors.
3. After replacing all the series or parallel groups you initially identified, you may find that more such groups reveal themselves. Replace those groups by using the same procedure as above until no more replacements are possible. If you then need to find the charge or potential difference for an individual original capacitor, you may have to retrace your steps.

EVALUATE *your answer:* Check whether your result makes sense. If the capacitors are connected in series, the equivalent capacitance C_{eq} must be *smaller* than any of the individual capacitances. If the capacitors are connected in parallel, C_{eq} must be *greater* than any of the individual capacitances.

EXAMPLE 24.5 Capacitors in series and in parallel

WITH VARIATION PROBLEMS

In Figs. 24.8 and 24.9, let $C_1 = 6.0 \mu\text{F}$, $C_2 = 3.0 \mu\text{F}$, and $V_{ab} = 18 \text{ V}$. Find the equivalent capacitance and the charge and potential difference for each capacitor when the capacitors are connected (a) in series (see Fig. 24.8) and (b) in parallel (see Fig. 24.9).

IDENTIFY and SET UP In both parts of this example a target variable is the equivalent capacitance C_{eq} , which is given by Eq. (24.5) for the series combination in part (a) and by Eq. (24.7) for the parallel combination in part (b). In each part we find the charge and potential difference from the definition of capacitance, Eq. (24.1), and the rules outlined in Problem-Solving Strategy 24.1.

EXECUTE (a) From Eq. (24.5) for a series combination,

$$\frac{1}{C_{\text{eq}}} = \frac{1}{C_1} + \frac{1}{C_2} = \frac{1}{6.0 \mu\text{F}} + \frac{1}{3.0 \mu\text{F}} \quad C_{\text{eq}} = 2.0 \mu\text{F}$$

The charge Q on each capacitor in series is the same as that on the equivalent capacitor:

$$Q = C_{\text{eq}}V = (2.0 \mu\text{F})(18 \text{ V}) = 36 \mu\text{C}$$

The potential difference across each capacitor is inversely proportional to its capacitance:

$$V_{ac} = V_1 = \frac{Q}{C_1} = \frac{36 \mu\text{C}}{6.0 \mu\text{F}} = 6.0 \text{ V}$$

$$V_{cb} = V_2 = \frac{Q}{C_2} = \frac{36 \mu\text{C}}{3.0 \mu\text{F}} = 12.0 \text{ V}$$

(b) From Eq. (24.7) for a parallel combination,

$$C_{\text{eq}} = C_1 + C_2 = 6.0 \mu\text{F} + 3.0 \mu\text{F} = 9.0 \mu\text{F}$$

The potential difference across each of the capacitors is the same as that across the equivalent capacitor, 18 V. The charge on each capacitor is directly proportional to its capacitance:

$$Q_1 = C_1V = (6.0 \mu\text{F})(18 \text{ V}) = 108 \mu\text{C}$$

$$Q_2 = C_2V = (3.0 \mu\text{F})(18 \text{ V}) = 54 \mu\text{C}$$

EVALUATE As expected, the equivalent capacitance C_{eq} for the series combination in part (a) is less than either C_1 or C_2 , while that for the parallel combination in part (b) is greater than either C_1 or C_2 . For two capacitors in series, as in part (a), the charge is the same on either capacitor and the *larger* potential difference appears across the capacitor with the *smaller* capacitance. Furthermore, the sum of the potential differences across the individual capacitors in series equals the potential difference across the equivalent capacitor: $V_{ac} + V_{cb} = V_{ab} = 18 \text{ V}$. By contrast, for two capacitors in parallel, as in part (b), each capacitor has the same potential difference and the *larger* charge appears on the capacitor with the *larger* capacitance. Can you show that the total

charge $Q_1 + Q_2$ on the parallel combination is equal to the charge $Q = C_{eq}V$ on the equivalent capacitor?

KEYCONCEPT When capacitors are combined in *series*, the charge is the same on each capacitor but the potential differences across the capacitors may not be the same. The reciprocal of the equivalent capacitance of a series combination equals the sum of the reciprocals of the individual capacitances. When capacitors are combined in *parallel*, the potential difference across each capacitor is the same but the charges on the capacitors may not be the same. The equivalent capacitance of a parallel combination equals the sum of the individual capacitances.

EXAMPLE 24.6 A capacitor network

WITH VARIATION PROBLEMS

Find the equivalent capacitance of the five-capacitor network shown in Fig. 24.10a.

IDENTIFY and SET UP These capacitors are neither all in series nor all in parallel. We can, however, identify portions of the arrangement that *are* either in series or parallel. We combine these as described in Problem-Solving Strategy 24.1 to find the net equivalent capacitance, using Eq. (24.5) for series connections and Eq. (24.7) for parallel connections.

EXECUTE The caption of Fig. 24.10 outlines our procedure. We first use Eq. (24.5) to replace the $12 \mu\text{F}$ and $6 \mu\text{F}$ series combination by its equivalent capacitance C' :

$$\frac{1}{C'} = \frac{1}{12 \mu\text{F}} + \frac{1}{6 \mu\text{F}} \quad C' = 4 \mu\text{F}$$

This gives us the equivalent combination of Fig. 24.10b. Now we see three capacitors in parallel, and we use Eq. (24.7) to replace them with their equivalent capacitance C'' :

$$C'' = 3 \mu\text{F} + 11 \mu\text{F} + 4 \mu\text{F} = 18 \mu\text{F}$$

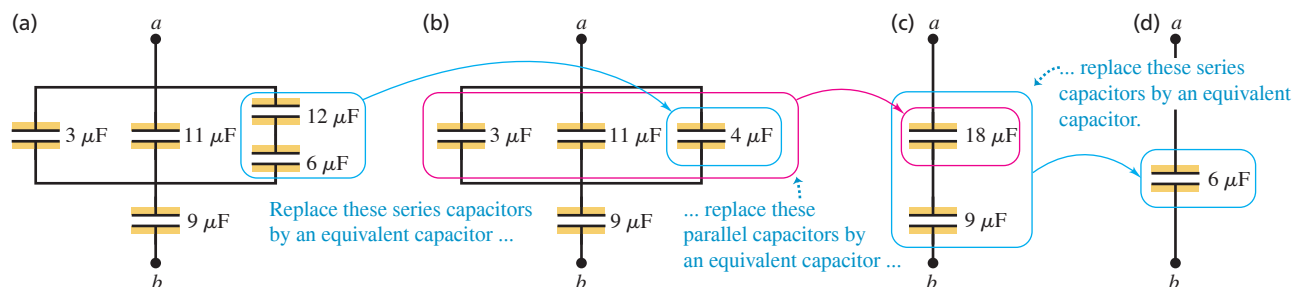
This gives us the equivalent combination of Fig. 24.10c, which has two capacitors in series. We use Eq. (24.5) to replace them with their equivalent capacitance C_{eq} , which is our target variable (Fig. 24.10d):

$$\frac{1}{C_{eq}} = \frac{1}{18 \mu\text{F}} + \frac{1}{9 \mu\text{F}} \quad C_{eq} = 6 \mu\text{F}$$

EVALUATE If the potential difference across the entire network in Fig. 24.10a is $V_{ab} = 9.0 \text{ V}$, the net charge on the network is $Q = C_{eq}V_{ab} = (6 \mu\text{F})(9.0 \text{ V}) = 54 \mu\text{C}$. Can you find the charge on, and the voltage across, each of the five individual capacitors?

KEYCONCEPT To analyze any network of capacitors, first identify which portions of the network are in series and which are in parallel. Then use the rules for series and parallel capacitors to determine the equivalent capacitance of the network.

Figure 24.10 (a) A capacitor network between points a and b . (b) The $12 \mu\text{F}$ and $6 \mu\text{F}$ capacitors in series in (a) are replaced by an equivalent $4 \mu\text{F}$ capacitor. (c) The $3 \mu\text{F}$, $11 \mu\text{F}$, and $4 \mu\text{F}$ capacitors in parallel in (b) are replaced by an equivalent $18 \mu\text{F}$ capacitor. (d) Finally, the $18 \mu\text{F}$ and $9 \mu\text{F}$ capacitors in series in (c) are replaced by an equivalent $6 \mu\text{F}$ capacitor.



TEST YOUR UNDERSTANDING OF SECTION 24.2 (a) How should you connect a $4 \mu\text{F}$ capacitor and an $8 \mu\text{F}$ capacitor so that the $4 \mu\text{F}$ capacitor has a greater potential difference across it than the $8 \mu\text{F}$ capacitor? (i) Series; (ii) parallel; (iii) either series or parallel; (iv) neither series nor parallel. (b) How should you connect them so that the $4 \mu\text{F}$ capacitor has a greater charge than the $8 \mu\text{F}$ capacitor? (i) Series; (ii) parallel; (iii) either series or parallel; (iv) neither series nor parallel.

ANSWER (i) In a series connection the two capacitors carry the same charge Q but have different potential differences $V_{ab} = Q/C$; the capacitor with the smaller capacitance C has the greater potential difference. In a parallel connection the two capacitors have the same potential difference V_{ab} but carry different charges $Q = CV_{ab}$; the capacitor with the larger capacitance C has the greater charge. Hence a $4 \mu\text{F}$ capacitor will have a greater potential difference than an $8 \mu\text{F}$ capacitor if the two are connected in series. The $4 \mu\text{F}$ capacitor cannot carry more charge than the $8 \mu\text{F}$ capacitor no matter how they are connected. In a series connection they will carry the same charge, and in a parallel connection the $8 \mu\text{F}$ capacitor will carry more charge.

24.3 ENERGY STORAGE IN CAPACITORS AND ELECTRIC-FIELD ENERGY

Many of the most important applications of capacitors depend on their ability to store energy. The electric potential energy stored in a charged capacitor is just equal to the amount of work required to charge it—that is, to separate opposite charges and place them on different conductors. When the capacitor is discharged, this stored energy is recovered as work done by electrical forces.

We can calculate the potential energy U of a charged capacitor by calculating the work W required to charge it. Suppose that when we are done charging the capacitor, the final charge is Q and the final potential difference is V . From Eq. (24.1) these quantities are related by

$$V = \frac{Q}{C}$$

Let q and v be the charge and potential difference, respectively, at an intermediate stage during the charging process; then $v = q/C$. At this stage the work dW required to transfer an additional element of charge dq is

$$dW = v dq = \frac{q dq}{C}$$

The total work W needed to increase the capacitor charge q from zero to Q is

$$W = \int_0^Q dW = \frac{1}{C} \int_0^Q q dq = \frac{Q^2}{2C} \quad (\text{work to charge a capacitor}) \quad (24.8)$$

This is also the total work done by the electric field on the charge when the capacitor discharges. Then q decreases from an initial value Q to zero as the elements of charge dq “fall” through potential differences v that vary from V down to zero.

If we define the potential energy of an *uncharged* capacitor to be zero, then W in Eq. (24.8) is equal to the potential energy U of the charged capacitor. The final stored charge is $Q = CV$, so we can express U (which is equal to W) as

$$\text{Potential energy stored in a capacitor} \quad U = \frac{Q^2}{2C} = \frac{1}{2} CV^2 = \frac{1}{2} QV \quad (24.9)$$

Magnitude of charge on each plate
Capacitance Potential difference between plates

When Q is in coulombs, C in farads (coulombs per volt), and V in volts (joules per coulomb), U is in joules.

The last form of Eq. (24.9), $U = \frac{1}{2} QV$, shows that the total work W required to charge the capacitor is equal to the total charge Q multiplied by the *average* potential difference $\frac{1}{2} V$ during the charging process.

The expression $U = \frac{1}{2} (Q^2/C)$ in Eq. (24.9) shows that a charged capacitor is the electrical analog of a stretched spring with elastic potential energy $U = \frac{1}{2} kx^2$. The charge Q is analogous to the elongation x , and the *reciprocal* of the capacitance, $1/C$, is analogous to the force constant k . The energy supplied to a capacitor in the charging process is analogous to the work we do on a spring when we stretch it.

Equations (24.8) and (24.9) tell us that capacitance measures the ability of a capacitor to store both energy and charge. If a capacitor is charged by connecting it to a battery or other source that provides a fixed potential difference V , then increasing the value of C gives a greater charge $Q = CV$ and a greater amount of stored energy $U = \frac{1}{2} CV^2$. If instead the goal is to transfer a given quantity of charge Q from one conductor to another, Eq. (24.8) shows that the work W required is inversely proportional to C ; the greater the capacitance, the easier it is to give a capacitor a fixed amount of charge.

Applications of Capacitors: Energy Storage

Most practical applications of capacitors take advantage of their ability to store and release energy. In an electronic flash unit used in photography, the energy stored in a capacitor (see Fig. 24.4) is released when the button is pressed to take a photograph. This provides a conducting path from one capacitor plate to the other through the flash tube. Once this path is established, the stored energy is rapidly converted into a brief but intense flash of light. An extreme example of the same principle is the Z machine at Sandia National Laboratories in New Mexico, which is used in experiments in controlled nuclear fusion (**Fig. 24.11**). A bank of charged capacitors releases more than a million joules of energy in just a few billionths of a second. For that brief space of time, the power output of the Z machine is 2.9×10^{14} W, or about 80 times the power output of all the electric power plants on earth combined!

In other applications, the energy is released more slowly. Springs in the suspension of an automobile help smooth out the ride by absorbing the energy from sudden jolts and releasing that energy gradually; in an analogous way, a capacitor in an electronic circuit can smooth out unwanted variations in voltage due to power surges. We'll discuss these circuits in detail in Chapter 26.

Electric-Field Energy

We can charge a capacitor by moving electrons directly from one plate to another. This requires doing work against the electric field between the plates. Thus we can think of the energy as being stored *in the field* in the region between the plates. To see this, let's find the energy *per unit volume* in the space between the plates of a parallel-plate capacitor with plate area A and separation d . We call this the **energy density**, denoted by u . From Eq. (24.9) the total stored potential energy is $\frac{1}{2}CV^2$ and the volume between the plates is Ad ; hence

$$u = \text{Energy density} = \frac{\frac{1}{2}CV^2}{Ad} \quad (24.10)$$

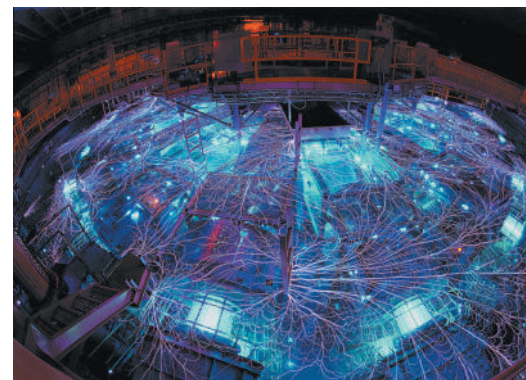
From Eq. (24.2) the capacitance C is given by $C = \epsilon_0 A/d$. The potential difference V is related to the electric-field magnitude E by $V = Ed$. If we use these expressions in Eq. (24.10), the geometric factors A and d cancel, and we find

$$\text{Electric energy density in a vacuum} \quad u = \frac{1}{2}\epsilon_0 E^2 \quad \text{Magnitude of electric field} \quad \text{Electric constant} \quad (24.11)$$

Although we have derived this relationship for a parallel-plate capacitor only, it turns out to be valid for any capacitor in vacuum and indeed *for any electric-field configuration in vacuum*. This result has an interesting implication. We think of vacuum as space with no matter in it, but vacuum can nevertheless have electric fields and therefore energy. Thus “empty” space need not be truly empty after all. We'll use this idea and Eq. (24.11) in Chapter 32 in connection with the energy transported by electromagnetic waves.

CAUTION **Electric-field energy is electric potential energy** It's a common misconception that electric-field energy is a new kind of energy, different from the electric potential energy described before. This is *not* the case; it is simply a different way of interpreting electric potential energy. We can regard the energy of a given system of charges as being a shared property of all the charges, or we can think of the energy as being a property of the electric field that the charges create. Either interpretation leads to the same value of the potential energy. ■

Figure 24.11 The Z machine uses a large number of capacitors in parallel to give a tremendous equivalent capacitance C (see Section 24.2). Hence a large amount of energy $U = \frac{1}{2}CV^2$ can be stored with even a modest potential difference V . The arcs shown here are produced when the capacitors discharge their energy into a target, which is no larger than a spool of thread. This heats the target to a temperature higher than 2×10^9 K.



EXAMPLE 24.7 Transferring charge and energy between capacitors**WITH VARIATION PROBLEMS**

We connect a capacitor $C_1 = 8.0 \mu\text{F}$ to a power supply, charge it to a potential difference $V_0 = 120 \text{ V}$, and disconnect the power supply (**Fig. 24.12**). Switch S is open. (a) What is the charge Q_0 on C_1 ? (b) What is the energy stored in C_1 ? (c) Capacitor $C_2 = 4.0 \mu\text{F}$ is initially uncharged. We close switch S . After charge no longer flows, what is the potential difference across each capacitor, and what is the charge on each capacitor? (d) What is the final energy of the system?

IDENTIFY and SET UP In parts (a) and (b) we find the charge Q_0 and stored energy U_{initial} for the single charged capacitor C_1 from Eqs. (24.1) and (24.9), respectively. After we close switch S , one wire connects the upper plates of the two capacitors and another wire connects the lower plates; the capacitors are now connected in parallel. In part (c) we use the character of the parallel connection to determine how Q_0 is shared between the two capacitors. In part (d) we again use Eq. (24.9) to find the energy stored in capacitors C_1 and C_2 ; the energy of the system is the sum of these values.

EXECUTE (a) The initial charge Q_0 on C_1 is

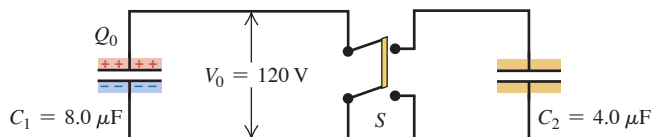
$$Q_0 = C_1 V_0 = (8.0 \mu\text{F})(120 \text{ V}) = 960 \mu\text{C}$$

(b) The energy initially stored in C_1 is

$$U_{\text{initial}} = \frac{1}{2} Q_0 V_0 = \frac{1}{2} (960 \times 10^{-6} \text{ C})(120 \text{ V}) = 0.058 \text{ J}$$

(c) When we close the switch, the positive charge Q_0 is distributed over the upper plates of both capacitors and the negative charge $-Q_0$ is distributed over the lower plates. Let Q_1 and Q_2 be the magnitudes of the final charges on the capacitors. Conservation of charge requires that $Q_1 + Q_2 = Q_0$. The potential difference V between the plates is the same for both capacitors because they are connected in parallel, so the charges are $Q_1 = C_1 V$ and $Q_2 = C_2 V$. We now have three independent equations relating the three unknowns Q_1 , Q_2 , and V . Solving these, we find

Figure 24.12 When the switch S is closed, the charged capacitor C_1 is connected to an uncharged capacitor C_2 . The center part of the switch is an insulating handle; charge can flow only between the two upper terminals and between the two lower terminals.



$$V = \frac{Q_0}{C_1 + C_2} = \frac{960 \mu\text{C}}{8.0 \mu\text{F} + 4.0 \mu\text{F}} = 80 \text{ V}$$

$$Q_1 = 640 \mu\text{C} \quad Q_2 = 320 \mu\text{C}$$

(d) The final energy of the system is

$$U_{\text{final}} = \frac{1}{2} Q_1 V + \frac{1}{2} Q_2 V = \frac{1}{2} Q_0 V \\ = \frac{1}{2} (960 \times 10^{-6} \text{ C})(80 \text{ V}) = 0.038 \text{ J}$$

EVALUATE The final energy is less than the initial energy; the difference was converted to energy of some other form. The conductors become a little warmer because of their resistance, and some energy is radiated as electromagnetic waves. We'll study the behavior of capacitors in more detail in Chapters 26 and 31.

KEYCONCEPT There are three useful ways to write the amount of electric potential energy stored in a charged capacitor: in terms of the charge Q on the plates and the potential difference V between the plates, in terms of Q and the capacitance C , and in terms of V and C .

EXAMPLE 24.8 Electric-field energy**WITH VARIATION PROBLEMS**

(a) What is the magnitude of the electric field required to store 1.00 J of electric potential energy in a volume of 1.00 m^3 in vacuum? (b) If the field magnitude is 10 times larger than that, how much energy is stored per cubic meter?

IDENTIFY and SET UP We use the relationship between the electric-field magnitude E and the energy density u . In part (a) we use the given information to find u ; then we use Eq. (24.11) to find the corresponding value of E . In part (b), Eq. (24.11) tells us how u varies with E .

EXECUTE (a) The desired energy density is $u = 1.00 \text{ J/m}^3$. Then from Eq. (24.11),

$$E = \sqrt{\frac{2u}{\epsilon_0}} = \sqrt{\frac{2(1.00 \text{ J/m}^3)}{8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2}} \\ = 4.75 \times 10^5 \text{ N/C} = 4.75 \times 10^5 \text{ V/m}$$

(b) Equation (24.11) shows that u is proportional to E^2 . If E increases by a factor of 10, u increases by a factor of $10^2 = 100$, so the energy density becomes $u = 100 \text{ J/m}^3$.

EVALUATE Dry air can sustain an electric field of about $3 \times 10^6 \text{ V/m}$ without experiencing *dielectric breakdown*, which we'll discuss in Section 24.4. There we'll see that field magnitudes in practical insulators can be even larger than this.

KEYCONCEPT Energy is stored in any region of space where an electric field is present. The electric energy density (energy per unit volume) at a point is proportional to the square of the electric-field magnitude at that point.

EXAMPLE 24.9 Two ways to calculate energy stored in a capacitor**WITH VARIATION PROBLEMS**

The spherical capacitor described in Example 24.3 (Section 24.1) has charges $+Q$ and $-Q$ on its inner and outer conductors. Find the electric potential energy stored in the capacitor (a) by using the capacitance C found in Example 24.3 and (b) by integrating the electric-field energy density u .

IDENTIFY and SET UP We can determine the energy U stored in a capacitor in two ways: in terms of the work done to put the charges on the two conductors, and in terms of the energy in the electric field between the conductors. The descriptions are equivalent, so they must give us the same result. In Example 24.3 we found the capacitance C and the field magnitude E in the space between the conductors. (The electric field is zero inside the inner sphere and is also zero outside the inner surface of the outer sphere, because a Gaussian surface with radius $r < r_a$ or $r > r_b$ encloses zero net charge. Hence the energy density is nonzero only in the space between the spheres, $r_a < r < r_b$.) In part (a) we use Eq. (24.9) to find U . In part (b) we use Eq. (24.11) to find u , which we integrate over the volume between the spheres to find U .

EXECUTE (a) From Example 24.3, the spherical capacitor has capacitance

$$C = 4\pi\epsilon_0 \frac{r_a r_b}{r_b - r_a}$$

where r_a and r_b are the radii of the inner and outer conducting spheres, respectively. From Eq. (24.9) the energy stored in this capacitor is

$$U = \frac{Q^2}{2C} = \frac{Q^2}{8\pi\epsilon_0} \frac{r_b - r_a}{r_a r_b}$$

(b) The electric field in the region $r_a < r < r_b$ between the two conducting spheres has magnitude $E = Q/4\pi\epsilon_0 r^2$. The energy density in this region is

$$u = \frac{1}{2}\epsilon_0 E^2 = \frac{1}{2}\epsilon_0 \left(\frac{Q}{4\pi\epsilon_0 r^2} \right)^2 = \frac{Q^2}{32\pi^2\epsilon_0 r^4}$$

The energy density is *not* uniform; it decreases rapidly with increasing distance from the center of the capacitor. To find the total electric-field energy, we integrate u (the energy per unit volume) over the region $r_a < r < r_b$. We divide this region into spherical shells of radius r , surface area $4\pi r^2$, thickness dr , and volume $dV = 4\pi r^2 dr$. Then

$$\begin{aligned} U &= \int u dV = \int_{r_a}^{r_b} \left(\frac{Q^2}{32\pi^2\epsilon_0 r^4} \right) 4\pi r^2 dr \\ &= \frac{Q^2}{8\pi\epsilon_0} \int_{r_a}^{r_b} \frac{dr}{r^2} = \frac{Q^2}{8\pi\epsilon_0} \left(-\frac{1}{r_b} + \frac{1}{r_a} \right) = \frac{Q^2}{8\pi\epsilon_0} \frac{r_b - r_a}{r_a r_b} \end{aligned}$$

EVALUATE Electric potential energy can be associated with either the *charges*, as in part (a), or the *field*, as in part (b); the calculated amount of stored energy is the same in either case.

KEYCONCEPT An alternative way to calculate the electric potential energy stored in a capacitor is to first find the electric field at each point in the space between the conducting plates. From this, find the electric energy density at each point. Finally, integrate the electric energy density over the entire volume where the electric field is present.

TEST YOUR UNDERSTANDING OF SECTION 24.3 You want to connect a $4\text{ }\mu\text{F}$ capacitor and an $8\text{ }\mu\text{F}$ capacitor. With which type of connection will the $4\text{ }\mu\text{F}$ capacitor have a greater amount of *stored energy* than the $8\text{ }\mu\text{F}$ capacitor? (i) Series; (ii) parallel; (iii) either series or parallel; (iv) neither series nor parallel.

ANSWER

(i) Capacitors connected in series carry the same charge Q . To compare the amount of energy stored, we use the expression $U = Q^2/2C$ from Eq. (24.9); it shows that the capacitor with the *smaller* capacitance ($C = 4\text{ }\mu\text{F}$) has more stored energy in a series combination. By contrast, capacitors in parallel have the same potential difference V , so to compare them we use $U = \frac{1}{2}CV^2$ from Eq. (24.9). It shows that in a parallel combination, the capacitor with the *larger* capacitance ($C = 8\text{ }\mu\text{F}$) has more stored energy. (If we had instead used $U = \frac{1}{2}CV^2$ to analyze the series combination, we would have to account for the different potential differences across the two capacitors. Likewise, using $U = Q^2/2C$ to study the parallel combination would require us to account for the different charges on the capacitors.)

24.4 DIELECTRICS

Most capacitors have a nonconducting material, or **dielectric**, between their conducting plates. A common type of capacitor uses long strips of metal foil for the plates, separated by strips of plastic sheet such as Mylar. A sandwich of these materials is rolled up, forming a unit that can provide a capacitance of several microfarads in a compact package (**Fig. 24.13**).

Placing a solid dielectric between the plates of a capacitor serves three functions. First, it solves the mechanical problem of maintaining two large metal sheets at a very small separation without actual contact.

Second, using a dielectric increases the maximum possible potential difference between the capacitor plates. Any insulating material, when subjected to a sufficiently large electric field, experiences a partial ionization that permits conduction through it (Section 23.3).

Figure 24.13 A common type of capacitor uses dielectric sheets to separate the conductors.

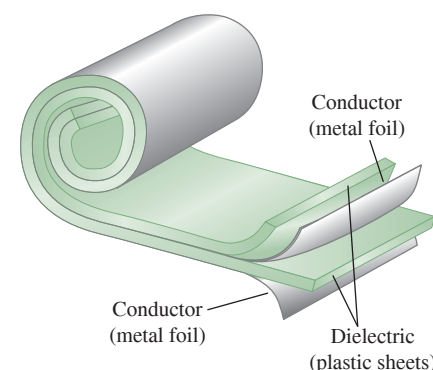
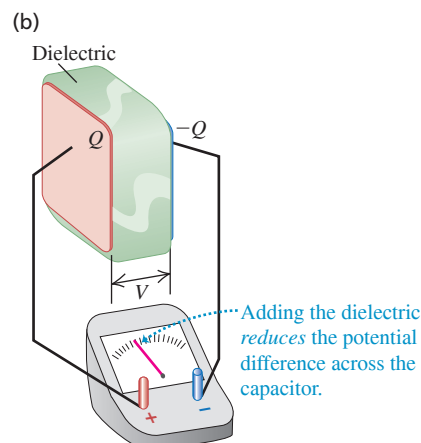
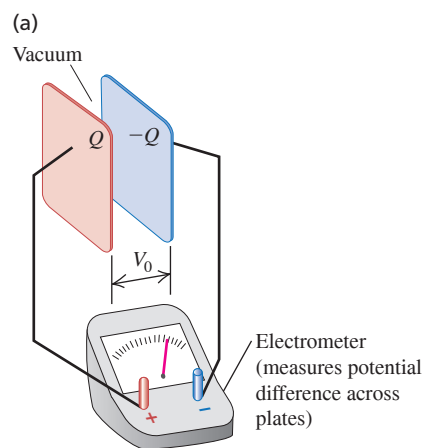


Figure 24.14 Effect of a dielectric between the plates of a parallel-plate capacitor. (a) With a given charge, the potential difference is V_0 . (b) With the same charge but with a dielectric between the plates, the potential difference V is smaller than V_0 .



This phenomenon is called **dielectric breakdown**. Many dielectric materials can tolerate stronger electric fields without breakdown than can air. Thus using a dielectric allows a capacitor to sustain a higher potential difference V and so store greater amounts of charge and energy.

Third, the capacitance of a capacitor of given dimensions is *greater* when there is a dielectric material between the plates than when there is vacuum. We can demonstrate this effect with the aid of a sensitive *electrometer*, a device that measures the potential difference between two conductors without letting any appreciable charge flow from one to the other. **Figure 24.14a** shows an electrometer connected across a charged capacitor, with magnitude of charge Q on each plate and potential difference V_0 . When we insert an uncharged sheet of dielectric, such as glass, paraffin, or polystyrene, between the plates, experiment shows that the potential difference *decreases* to a smaller value V (Fig. 24.14b). When we remove the dielectric, the potential difference returns to its original value V_0 , showing that the original charges on the plates have not changed.

The original capacitance C_0 is given by $C_0 = Q/V_0$, and the capacitance C with the dielectric present is $C = Q/V$. The charge Q is the same in both cases, and V is less than V_0 , so we conclude that the capacitance C with the dielectric present is *greater* than C_0 . When the space between plates is completely filled by the dielectric, the ratio of C to C_0 (equal to the ratio of V_0 to V) is called the **dielectric constant** of the material, K :

$$K = \frac{C}{C_0} \quad (\text{definition of dielectric constant}) \quad (24.12)$$

When the charge is constant, $Q = C_0 V_0 = CV$ and $C/C_0 = V_0/V$. In this case,

$$V = \frac{V_0}{K} \quad (\text{when } Q \text{ is constant}) \quad (24.13)$$

With the dielectric present, the potential difference for a given charge Q is *reduced* by a factor K .

The dielectric constant K is a pure number. Because C is always greater than C_0 , K is always greater than unity. **Table 24.1** gives some representative values of K . For vacuum, $K = 1$ by definition. For air at ordinary temperatures and pressures, K is about 1.0006; this is so nearly equal to 1 that for most purposes an air capacitor is equivalent to one in vacuum. Note that while water has a very large value of K , it is usually not a very practical dielectric for use in capacitors. The reason is that while pure water is a very poor conductor, it is also an excellent ionic solvent. Any ions that are dissolved in the water will cause charge to flow between the capacitor plates, so the capacitor discharges.

CAUTION Dielectric constant vs. electric constant Don't confuse the *dielectric* constant K with the *electric* constant ϵ_0 . The value of K is a pure number with no units and is different for different materials (see Table 24.1). By contrast, ϵ_0 is a universal constant with units $C^2/N \cdot m^2$ or F/m . ■

TABLE 24.1 Values of Dielectric Constant K at 20°C

Material	K	Material	K
Vacuum	1	Polyvinyl chloride	3.18
Air (1 atm)	1.00059	Plexiglas®	3.40
Air (100 atm)	1.0548	Glass	5–10
Teflon®	2.1	Neoprene	6.70
Polyethylene	2.25	Germanium	16
Benzene	2.28	Glycerin	42.5
Mica	3–6	Water	80.4
Mylar®	3.1	Strontium titanate	310

No real dielectric is a perfect insulator. Hence there is always some *leakage current* between the charged plates of a capacitor with a dielectric. We tacitly ignored this effect in Section 24.2 when we derived expressions for the equivalent capacitances of capacitors in series, Eq. (24.5), and in parallel, Eq. (24.7). But if a leakage current flows for a long enough time to substantially change the charges from the values we used to derive Eqs. (24.5) and (24.7), those equations may no longer be accurate.

Induced Charge and Polarization

When a dielectric material is inserted between the plates while the charge is kept constant, the potential difference between the plates decreases by a factor K . Therefore the electric field between the plates must decrease by the same factor. If E_0 is the vacuum value and E is the value with the dielectric, then

$$E = \frac{E_0}{K} \quad (\text{when } Q \text{ is constant}) \quad (24.14)$$

Since the electric-field magnitude is smaller when the dielectric is present, the surface charge density (which causes the field) must be smaller as well. The surface charge on the conducting plates does not change, but an *induced* charge of the opposite sign appears on each surface of the dielectric (Fig. 24.15). The dielectric was originally electrically neutral and is still neutral; the induced surface charges arise as a result of *redistribution* of positive and negative charges within the dielectric material, a phenomenon called **polarization**. We first encountered polarization in Section 21.2, and we suggest that you reread the discussion of Fig. 21.8. We'll assume that the induced surface charge is *directly proportional* to the electric-field magnitude E in the material; this is indeed the case for many common dielectrics. (This direct proportionality is analogous to Hooke's law for a spring.) In that case, K is a constant for any particular material. When the electric field is very strong or if the dielectric is made of certain crystalline materials, the relationship between induced charge and the electric field can be more complex; we won't consider such cases here.

We can derive a relationship between this induced surface charge and the charge on the plates. Let σ_i be the magnitude of the charge per unit area induced on the surfaces of the dielectric (the induced surface charge density). The magnitude of the surface charge density on the capacitor plates is σ , as usual. Then the *net* surface charge on each side of the capacitor has magnitude $(\sigma - \sigma_i)$; see Fig. 24.15b. As we found in Example 21.12 (Section 21.5) and in Example 22.8 (Section 22.4), the field between the plates is related to the net surface charge density by $E = \sigma_{\text{net}}/\epsilon_0$. Without and with the dielectric, respectively,

$$E_0 = \frac{\sigma}{\epsilon_0} \quad E = \frac{\sigma - \sigma_i}{\epsilon_0} \quad (24.15)$$

Using these expressions in Eq. (24.14) and rearranging the result, we find

$$\sigma_i = \sigma \left(1 - \frac{1}{K} \right) \quad (\text{induced surface charge density}) \quad (24.16)$$

This equation shows that when K is very large, σ_i is nearly as large as σ . In this case, σ_i nearly cancels σ , and the field and potential difference are much smaller than their values in vacuum.

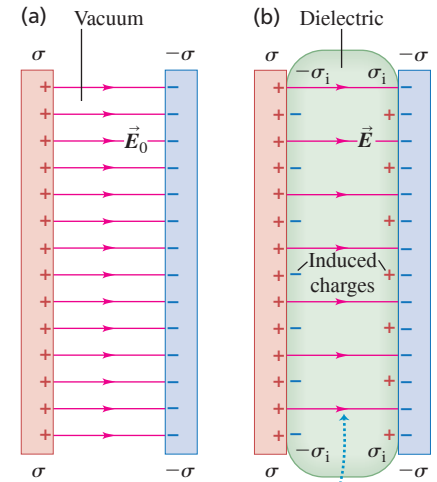
The product $K\epsilon_0$ is called the **permittivity** of the dielectric, denoted by ϵ :

$$\epsilon = K\epsilon_0 \quad (\text{definition of permittivity}) \quad (24.17)$$

In terms of ϵ we can express the electric field within the dielectric as

$$E = \frac{\sigma}{\epsilon} \quad (24.18)$$

Figure 24.15 Electric field lines with (a) vacuum between the plates and (b) dielectric between the plates.



For a given charge density σ , the induced charges on the dielectric's surfaces reduce the electric field between the plates.

APPLICATION Capacitors in the Toolbox

Several practical devices rely on the way a capacitor responds to a change in dielectric constant. One example is an electric stud finder, used to locate metal segments hidden behind a wall's surface. It consists of a metal plate with associated circuitry. The plate acts as one half of a capacitor; the wall acts as the other half. If the stud finder moves over a metal segment, the effective dielectric constant for the capacitor changes, which changes the capacitance and triggers a signal.



Then

$$C = KC_0 = K\epsilon_0 \frac{A}{d} = \epsilon \frac{A}{d} \quad (24.19)$$

Capacitance of a parallel-plate capacitor, dielectric between plates: C
 Dielectric constant: K
 Area of each plate: A
 Distance between plates: d
 Capacitance without dielectric: C_0
 Electric constant: ϵ_0
 Permittivity: $\epsilon = K\epsilon_0$

We can repeat the derivation of Eq. (24.11) for the energy density u in an electric field for the case in which a dielectric is present. The result is

$$u = \frac{1}{2}K\epsilon_0 E^2 = \frac{1}{2}\epsilon E^2 \quad (24.20)$$

Electric energy density in a dielectric: u
 Dielectric constant: K
 Electric constant: ϵ_0
 Permittivity: $\epsilon = K\epsilon_0$
 Magnitude of electric field: E

CAUTION Capacitance is independent of charge or voltage

Equation (24.19) shows that the capacitance C of a capacitor depends on only the capacitor geometry (the size, shape, and position of its plates) and whether or not a dielectric is present. The capacitance does *not* depend on the amount of charge placed on the plates or on the voltage between the plates!

In empty space, where $K = 1$, $\epsilon = \epsilon_0$ and Eqs. (24.19) and (24.20) reduce to Eqs. (24.2) and (24.11), respectively, for a parallel-plate capacitor in vacuum. For this reason, ϵ_0 is sometimes called the “permittivity of free space” or the “permittivity of vacuum.” Because K is a pure number, ϵ and ϵ_0 have the same units, $C^2/N \cdot m^2$ or F/m .

Equation (24.19) shows that extremely high capacitances can be obtained with plates that have a large surface area A and are separated by a small distance d by a dielectric with a large value of K . In an *electrolytic double-layer capacitor*, tiny carbon granules adhere to each plate: A is the combined surface area of the granules, which can be tremendous. The plates with granules attached are separated by a very thin dielectric sheet. Such a capacitor can have a capacitance of 5000 F yet fit in the palm of your hand (compare Example 24.1 in Section 24.1).

PROBLEM-SOLVING STRATEGY 24.2 Dielectrics

IDENTIFY the relevant concepts: The relationships in this section are useful whenever there is an electric field in a dielectric, such as a dielectric between charged capacitor plates. Typically you must relate the potential difference V_{ab} between the plates, the electric-field magnitude E in the capacitor, the charge density σ on the capacitor plates, and the induced charge density σ_i on the surfaces of the capacitor.

SET UP the problem using the following steps:

1. Make a drawing of the situation.
2. Identify the target variables, and choose which equations from this section will help you solve for those variables.

EXECUTE the solution as follows:

1. In problems such as the next example, it is easy to get lost in a blizzard of formulas. Ask yourself at each step what kind of quantity each symbol represents. For example, distinguish clearly

between charges and charge densities, and between electric fields and electric potential differences.

2. Check for consistency of units. Distances must be in meters. A microfarad is 10^{-6} farad, and so on. Don't confuse the numerical value of ϵ_0 with the value of $1/4\pi\epsilon_0$. Electric-field magnitude can be expressed in both N/C and V/m . The units of ϵ_0 are $C^2/N \cdot m^2$ or F/m .

EVALUATE your answer: With a dielectric present, (a) the capacitance is greater than without a dielectric; (b) for a given charge on the capacitor, the electric field and potential difference are less than without a dielectric; and (c) the magnitude of the induced surface charge density σ_i on the dielectric is less than that of the charge density σ on the capacitor plates.

EXAMPLE 24.10 A capacitor with and without a dielectric

WITH VARIATION PROBLEMS

Suppose the parallel plates in Fig. 24.15 each have an area of 2000 cm^2 ($2.00 \times 10^{-1} \text{ m}^2$) and are 1.00 cm ($1.00 \times 10^{-2} \text{ m}$) apart. We connect the capacitor to a power supply, charge it to a potential difference $V_0 = 3.00 \text{ kV}$, and disconnect the power supply. We then insert a sheet of insulating plastic material between the plates, completely filling the space between them. We find that the potential difference decreases to 1.00 kV while the charge on each capacitor

plate remains constant. Find (a) the original capacitance C_0 ; (b) the magnitude of charge Q on each plate; (c) the capacitance C after the dielectric is inserted; (d) the dielectric constant K of the dielectric; (e) the permittivity ϵ of the dielectric; (f) the magnitude of the induced charge Q_i on each face of the dielectric; (g) the original electric field E_0 between the plates; and (h) the electric field E after the dielectric is inserted.

IDENTIFY and SET UP This problem uses most of the relationships we have discussed for capacitors and dielectrics. (Energy relationships are treated in Example 24.11.) Most of the target variables can be obtained in several ways. The methods used below are a sample; we encourage you to think of others and compare your results.

EXECUTE (a) With vacuum between the plates, we use Eq. (24.19) with $K = 1$:

$$C_0 = \epsilon_0 \frac{A}{d} = (8.85 \times 10^{-12} \text{ F/m}) \frac{2.00 \times 10^{-1} \text{ m}^2}{1.00 \times 10^{-2} \text{ m}} \\ = 1.77 \times 10^{-10} \text{ F} = 177 \text{ pF}$$

(b) From the definition of capacitance, Eq. (24.1),

$$Q = C_0 V_0 = (1.77 \times 10^{-10} \text{ F})(3.00 \times 10^3 \text{ V}) \\ = 5.31 \times 10^{-7} \text{ C} = 0.531 \mu\text{C}$$

(c) When the dielectric is inserted, Q is unchanged but the potential difference decreases to $V = 1.00 \text{ kV}$. Hence from Eq. (24.1), the new capacitance is

$$C = \frac{Q}{V} = \frac{5.31 \times 10^{-7} \text{ C}}{1.00 \times 10^3 \text{ V}} = 5.31 \times 10^{-10} \text{ F} = 531 \text{ pF}$$

(d) From Eq. (24.12), the dielectric constant is

$$K = \frac{C}{C_0} = \frac{5.31 \times 10^{-10} \text{ F}}{1.77 \times 10^{-10} \text{ F}} = \frac{531 \text{ pF}}{177 \text{ pF}} = 3.00$$

Alternatively, from Eq. (24.13),

$$K = \frac{V_0}{V} = \frac{3000 \text{ V}}{1000 \text{ V}} = 3.00$$

(e) With K from part (d) in Eq. (24.17), the permittivity is

$$\epsilon = K\epsilon_0 = (3.00)(8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2) = 2.66 \times 10^{-11} \text{ C}^2/\text{N} \cdot \text{m}^2$$

(f) Multiplying both sides of Eq. (24.16) by the plate area A gives the induced charge $Q_i = \sigma_i A$ in terms of the charge $Q = \sigma A$ on each plate:

$$Q_i = Q \left(1 - \frac{1}{K} \right) = (5.31 \times 10^{-7} \text{ C}) \left(1 - \frac{1}{3.00} \right) = 3.54 \times 10^{-7} \text{ C}$$

(g) Since the electric field between the plates is uniform, its magnitude is the potential difference divided by the plate separation:

$$E_0 = \frac{V_0}{d} = \frac{3000 \text{ V}}{1.00 \times 10^{-2} \text{ m}} = 3.00 \times 10^5 \text{ V/m}$$

(h) After the dielectric is inserted,

$$E = \frac{V}{d} = \frac{1000 \text{ V}}{1.00 \times 10^{-2} \text{ m}} = 1.00 \times 10^5 \text{ V/m}$$

or, from Eq. (24.18),

$$E = \frac{\sigma}{\epsilon} = \frac{Q}{\epsilon A} = \frac{5.31 \times 10^{-7} \text{ C}}{(2.66 \times 10^{-11} \text{ C}^2/\text{N} \cdot \text{m}^2)(2.00 \times 10^{-1} \text{ m}^2)} \\ = 1.00 \times 10^5 \text{ V/m}$$

or, from Eq. (24.15),

$$E = \frac{\sigma - \sigma_i}{\epsilon_0} = \frac{Q - Q_i}{\epsilon_0 A} \\ = \frac{(5.31 - 3.54) \times 10^{-7} \text{ C}}{(8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)(2.00 \times 10^{-1} \text{ m}^2)} \\ = 1.00 \times 10^5 \text{ V/m}$$

or, from Eq. (24.14),

$$E = \frac{E_0}{K} = \frac{3.00 \times 10^5 \text{ V/m}}{3.00} = 1.00 \times 10^5 \text{ V/m}$$

EVALUATE Inserting the dielectric increased the capacitance by a factor of $K = 3.00$ and reduced the electric field between the plates by a factor of $1/K = 1/3.00$. It did so by developing induced charges on the faces of the dielectric of magnitude $Q(1 - 1/K) = Q(1 - 1/3.00) = 0.667Q$.

KEYCONCEPT Adding a dielectric (with dielectric constant K) that fills the space between the plates of a capacitor increases the capacitance by a factor of K . For a given amount of charge, adding the dielectric also reduces both the potential difference between the plates and the electric field at each point by a factor of $1/K$.

EXAMPLE 24.11 Energy storage with and without a dielectric

WITH VARIATION PROBLEMS

Find the energy stored in the electric field of the capacitor in Example 24.10 and the energy density, both before and after the dielectric sheet is inserted.

IDENTIFY and SET UP We consider the ideas of energy stored in a capacitor and of electric-field energy density. We use Eq. (24.9) to find the stored energy and Eq. (24.20) to find the energy density.

EXECUTE From Eq. (24.9), the stored energies U_0 and U without and with the dielectric in place are

$$U_0 = \frac{1}{2} C_0 V_0^2 = \frac{1}{2} (1.77 \times 10^{-10} \text{ F})(3000 \text{ V})^2 = 7.97 \times 10^{-4} \text{ J} \\ U = \frac{1}{2} C V^2 = \frac{1}{2} (5.31 \times 10^{-10} \text{ F})(1000 \text{ V})^2 = 2.66 \times 10^{-4} \text{ J}$$

The final energy is one-third of the original energy.

Equation (24.20) gives the energy densities without and with the dielectric:

$$u_0 = \frac{1}{2} \epsilon_0 E_0^2 = \frac{1}{2} (8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)(3.00 \times 10^5 \text{ N/C})^2 \\ = 0.398 \text{ J/m}^3 \\ u = \frac{1}{2} \epsilon E^2 = \frac{1}{2} (2.66 \times 10^{-11} \text{ C}^2/\text{N} \cdot \text{m}^2)(1.00 \times 10^5 \text{ N/C})^2 \\ = 0.133 \text{ J/m}^3$$

The energy density with the dielectric is one-third of the original energy density.

Continued

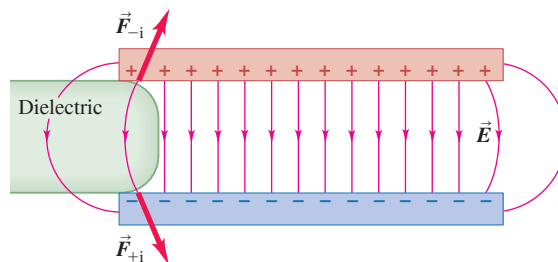
EVALUATE We can check our answer for u_0 by noting that the volume between the plates is $V_{\text{between}} = (0.200 \text{ m}^2)(0.0100 \text{ m}) = 0.00200 \text{ m}^3$. Since the electric field between the plates is uniform, u_0 is uniform as well and the energy density is just the stored energy divided by the volume:

$$u_0 = \frac{U_0}{V_{\text{between}}} = \frac{7.97 \times 10^{-4} \text{ J}}{0.00200 \text{ m}^3} = 0.398 \text{ J/m}^3$$

This agrees with our earlier answer. You can use the same approach to check our result for u .

In general, when a dielectric is inserted into a capacitor while the charge on each plate remains the same, the permittivity ϵ increases by a factor of K (the dielectric constant), and the electric field E and the energy density $u = \frac{1}{2}\epsilon E^2$ decrease by a factor of $1/K$. Where does the energy go? The answer lies in the fringing field at the edges of a real parallel-plate capacitor. As **Fig. 24.16** shows, that field tends to pull the dielectric into the space between the plates, doing work on it as it does so. We could attach a spring to the left end of the dielectric in Fig. 24.16 and use this force to stretch the spring. Because work is done by the field, the field energy density decreases.

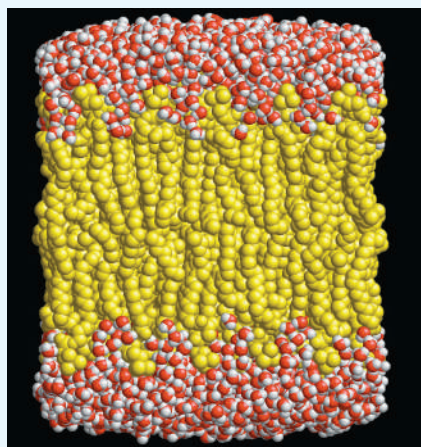
Figure 24.16 The fringing field at the edges of the capacitor exerts forces $\vec{F}_{-i}$ and $\vec{F}_{+i}$ on the negative and positive induced surface charges of a dielectric, pulling the dielectric into the capacitor.



KEYCONCEPT Adding a dielectric (with dielectric constant K) that fills the space between the plates of a capacitor reduces the electric field, the electric energy density, and the total stored energy, all by a factor of $1/K$.

BIO APPLICATION Dielectric Cell Membrane

The membrane of a living cell behaves like a dielectric between the plates of a capacitor. The membrane is made of two sheets of lipid molecules, with their water-insoluble ends in the middle and their water-soluble ends (shown in red) on the outer surfaces. Conductive fluids on either side of the membrane (water with negative ions inside the cell, water with positive ions outside) act as charged capacitor plates, and the nonconducting membrane acts as a dielectric with K of about 10. The potential difference V across the membrane is about 0.07 V and the membrane thickness d is about $7 \times 10^{-9} \text{ m}$, so the electric field $E = V/d$ in the membrane is about 10^7 V/m —close to the dielectric strength of the membrane. If the membrane were made of air, V and E would be larger by a factor of $K \approx 10$ and dielectric breakdown would occur.



Dielectric Breakdown

We mentioned earlier that when a dielectric is subjected to a sufficiently strong electric field, *dielectric breakdown* takes place and the dielectric becomes a conductor. This occurs when the electric field is so strong that electrons are ripped loose from their molecules and crash into other molecules, liberating even more electrons. This avalanche of moving charge forms a spark or arc discharge. Lightning is a dramatic example of dielectric breakdown in air.

Because of dielectric breakdown, capacitors always have maximum voltage ratings. When a capacitor is subjected to excessive voltage, an arc may form through a layer of dielectric, burning or melting a hole in it. This arc creates a conducting path (a short circuit) between the conductors. If a conducting path remains after the arc is extinguished, the device is rendered permanently useless as a capacitor.

The maximum electric-field magnitude that a material can withstand without the occurrence of breakdown is called its **dielectric strength**. This quantity is affected significantly by temperature, trace impurities, small irregularities in the metal electrodes, and other factors that are difficult to control. For this reason we can give only approximate figures for dielectric strengths. The dielectric strength of dry air is about $3 \times 10^6 \text{ V/m}$. **Table 24.2** lists the dielectric strengths of a few common insulating materials. All of the values are substantially greater than the value for air. For example, a layer of polycarbonate 0.01 mm thick (about the smallest practical thickness) has 10 times the dielectric strength of air and can withstand a maximum voltage of about $(3 \times 10^7 \text{ V/m})(1 \times 10^{-5} \text{ m}) = 300 \text{ V}$.

TABLE 24.2 Dielectric Constant and Dielectric Strength of Some Insulating Materials

Material	Dielectric Constant, K	Dielectric Strength, E_m (V/m)
Polycarbonate	2.8	3×10^7
Polyester	3.3	6×10^7
Polypropylene	2.2	7×10^7
Polystyrene	2.6	2×10^7
Pyrex® glass	4.7	1×10^7

TEST YOUR UNDERSTANDING OF SECTION 24.4 The space between the plates of an isolated parallel-plate capacitor is filled by a slab of dielectric with dielectric constant K . The two plates of the capacitor have charges Q and $-Q$. You pull out the dielectric slab. If the charges do not change, how does the energy in the capacitor change when you remove the slab? (i) It increases; (ii) it decreases; (iii) it remains the same.

ANSWER

(i) Here Q remains the same, so we use $U = Q^2/2C$ from Eq. (24.9). Removing the dielectric lowers the capacitance by a factor of $1/K$; since U is inversely proportional to C , the stored energy *increases* by a factor of K . It takes work to pull the dielectric slab out of the capacitor because the fringing field tries to pull the slab back in (Fig. 24.16). The work that you do goes into the energy stored in the capacitor.

24.5 MOLECULAR MODEL OF INDUCED CHARGE

In Section 24.4 we discussed induced surface charges on a dielectric in an electric field. Now let's look at how these surface charges can arise. If the material were a *conductor*, the answer would be simple. Conductors contain charge that is free to move, and when an electric field is present, some of the charge redistributes itself on the surface so that there is no electric field inside the conductor. But an ideal dielectric has *no* charges that are free to move, so how can a surface charge occur?

To understand this, we have to look again at rearrangement of charge at the *molecular* level. Some molecules, such as H_2O and N_2O , have equal amounts of positive and negative charges but a lopsided distribution, with excess positive charge concentrated on one side of the molecule and negative charge on the other. As we described in Section 21.7, such an arrangement is called an *electric dipole*, and the molecule is called a *polar molecule*. When no electric field is present in a gas or liquid with polar molecules, the molecules are oriented randomly (Fig. 24.17a). In an electric field, however, they tend to orient themselves as in Fig. 24.17b, as a result of the electric-field torques described in Section 21.7. Because of thermal agitation, the alignment of the molecules with $\vec{E}$ is not perfect.

Even a molecule that is *not* ordinarily polar *becomes* a dipole when it is placed in an electric field because the field pushes the positive charges in the molecules in the direction of the field and pushes the negative charges in the opposite direction. This causes a redistribution of charge within the molecule (Fig. 24.18). Such dipoles are called *induced* dipoles.

With either polar or nonpolar molecules, the redistribution of charge caused by the field leads to the formation of a layer of charge on each surface of the dielectric material (Fig. 24.19). These layers are the surface charges described in Section 24.4; their surface charge density is denoted by σ_i . The charges are *not* free to move indefinitely, as they

Figure 24.19 Polarization of a dielectric in an electric field $\vec{E}$ gives rise to thin layers of bound charges on the surfaces, creating surface charge densities σ_i and $-\sigma_i$. The sizes of the molecules are greatly exaggerated for clarity.

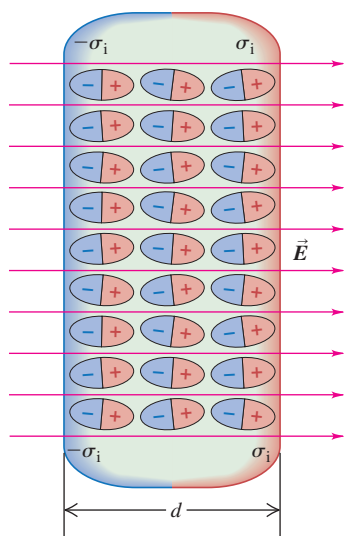


Figure 24.17 Polar molecules (a) without and (b) with an applied electric field $\vec{E}$.

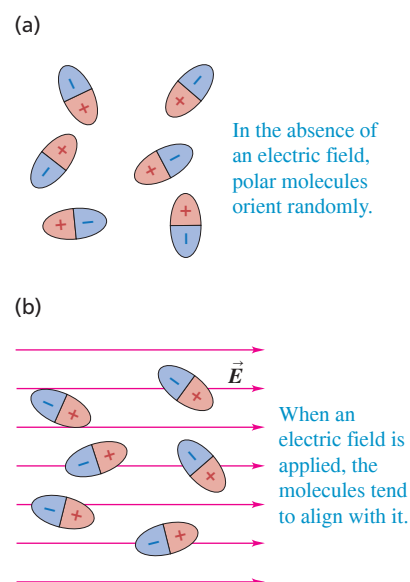
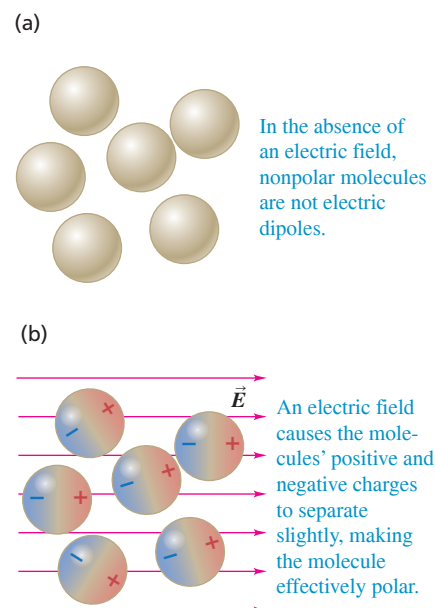


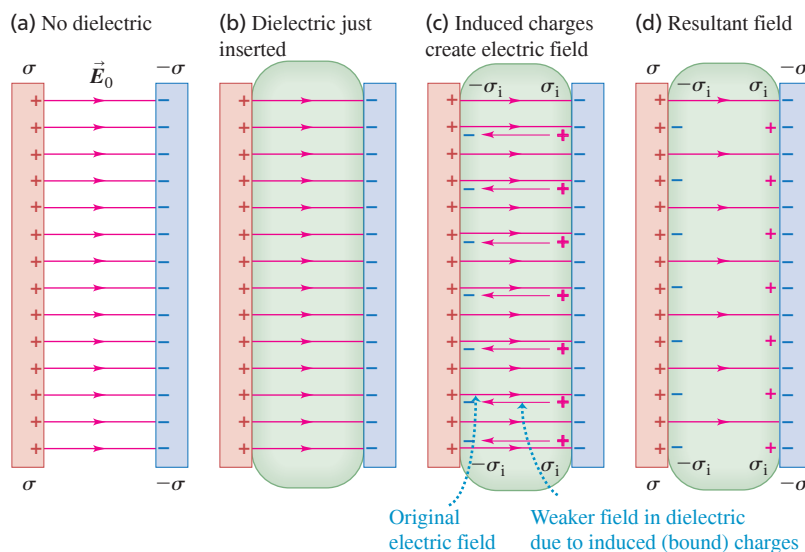
Figure 24.18 Nonpolar molecules (a) without and (b) with an applied electric field $\vec{E}$.



APPLICATION Smartphones, Capacitors, and Dielectrics A typical smartphone contains several hundred miniature capacitors, each of which uses a ceramic dielectric between its plates to boost its capacitance. These capacitors are used in every subsystem of the smartphone, including the main processor, the dynamic random access memory, the on-off and volume controls, the microphone, and the touch screen (see page 787).



Figure 24.20 (a) Electric field of magnitude E_0 between two charged plates. (b) Introduction of a dielectric of dielectric constant K . (c) The induced surface charges and their field. (d) Resultant field of magnitude E_0/K .

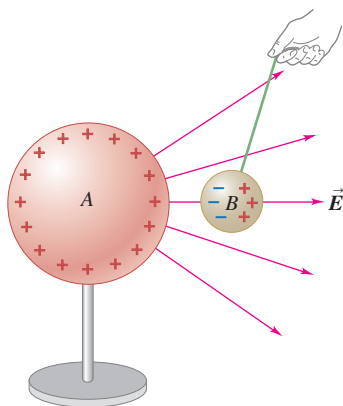


would be in a conductor, because each charge is bound to a molecule. They are in fact called **bound charges** to distinguish them from the **free charges** that are added to and removed from the conducting capacitor plates. In the interior of the material the net charge per unit volume remains zero. As we have seen, this redistribution of charge is called *polarization*, and we say that the material is *polarized*.

The four parts of Fig. 24.20 show the behavior of a slab of dielectric when it is inserted in the field between a pair of oppositely charged capacitor plates. Figure 24.20a shows the original field. Figure 24.20b is the situation after the dielectric has been inserted but before any rearrangement of charges has occurred. Figure 24.20c shows by thinner arrows the additional field set up in the dielectric by its induced surface charges. This field is *opposite* to the original field, but it is not great enough to cancel the original field completely because the charges in the dielectric are not free to move indefinitely. The resultant field in the dielectric, shown in Fig. 24.20d, is therefore decreased in magnitude. In the field-line representation, some of the field lines leaving the positive plate go through the dielectric, while others terminate on the induced charges on the faces of the dielectric.

As we discussed in Section 21.2, polarization is also the reason a charged object, such as an electrified plastic rod, can exert a force on an *uncharged* object such as a bit of paper or a pith ball. **Figure 24.21** shows an uncharged dielectric sphere B in the radial field of a positively charged object A . The induced positive charges on B experience a force toward the right, while the force on the induced negative charges is toward the left. The negative charges are closer to A , and thus are in a stronger field, than are the positive charges. The force toward the left is stronger than that toward the right, and B is attracted toward A , even though its net charge is zero. The attraction occurs whether the sign of A 's charge is positive or negative (see Fig. 21.8). Furthermore, the effect is not limited to dielectrics; an uncharged conducting object would be attracted in the same way.

Figure 24.21 A neutral sphere B in the radial electric field of a positively charged sphere A is attracted to the charge because of polarization.



TEST YOUR UNDERSTANDING OF SECTION 24.5 A parallel-plate capacitor has charges Q and $-Q$ on its two plates. A dielectric slab with $K = 3$ is then inserted into the space between the plates as shown in Fig. 24.20. Rank the following electric-field magnitudes in order from largest to smallest. (i) The field before the slab is inserted; (ii) the resultant field after the slab is inserted; (iii) the field due to the bound charges.

ANSWER

(i), (iii), (ii) Equation (24.14) says that if E_0 is the initial electric-field magnitude (before the dielectric slab is inserted), then the resultant field magnitude after the slab is inserted is $E_0/K = E_0/3$. The magnitude of the resultant field equals the difference between the initial field magnitude and the magnitude of the field due to the bound charges (see Fig. 24.20). Hence $E_0 - E_i = E_0/3$ and $E_i = 2E_0/3$.

24.6 GAUSS'S LAW IN DIELECTRICS

We can extend the analysis of Section 24.4 to reformulate Gauss's law in a form that is particularly useful for dielectrics. **Figure 24.22** is a close-up view of the left capacitor plate and left surface of the dielectric in Fig. 24.15b. Let's apply Gauss's law to the rectangular box shown in cross section by the purple line; the surface area of the left and right sides is A . The left side is embedded in the conductor that forms the left capacitor plate, and so the electric field everywhere on that surface is zero. The right side is embedded in the dielectric, where the electric field has magnitude E , and $E_{\perp} = 0$ everywhere on the other four sides. The total charge enclosed, including both the charge on the capacitor plate and the induced charge on the dielectric surface, is $Q_{\text{encl}} = (\sigma - \sigma_i)A$, so Gauss's law gives

$$EA = \frac{(\sigma - \sigma_i)A}{\epsilon_0} \quad (24.21)$$

This equation is not very illuminating as it stands because it relates two unknown quantities: E inside the dielectric and the induced surface charge density σ_i . But now we can use Eq. (24.16), developed for this same situation, to simplify this equation by eliminating σ_i . Equation (24.16) is

$$\sigma_i = \sigma \left(1 - \frac{1}{K}\right)$$

or

$$\sigma - \sigma_i = \frac{\sigma}{K}$$

Combining this with Eq. (24.21), we get

$$EA = \frac{\sigma A}{K\epsilon_0}$$

or

$$KEA = \frac{\sigma A}{\epsilon_0} \quad (24.22)$$

Equation (24.22) says that the flux of $K\vec{E}$, not $\vec{E}$, through the Gaussian surface in Fig. 24.22 is equal to the enclosed *free* charge σA divided by ϵ_0 . It turns out that for *any* Gaussian surface, whenever the induced charge is proportional to the electric field in the material, we can rewrite Gauss's law as

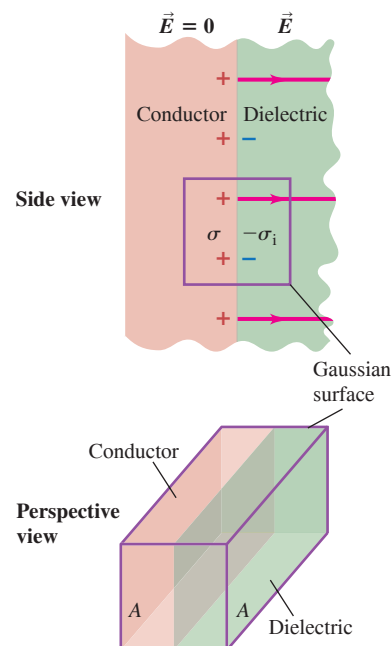
$$\oint K\vec{E} \cdot d\vec{A} = \frac{Q_{\text{encl-free}}}{\epsilon_0} \quad (24.23)$$

Gauss's law in a dielectric:
Dielectric constant
Total free charge enclosed by surface
Electric constant

Surface integral of $K\vec{E}$ over a closed surface

where $Q_{\text{encl-free}}$ is the total *free* charge (not bound charge) enclosed by the Gaussian surface. The significance of these results is that the right sides contain only the *free* charge on the conductor, not the bound (induced) charge. In fact, although we have not proved it, Eq. (24.23) remains valid even when different parts of the Gaussian surface are embedded in dielectrics having different values of K , provided that the value of K in each dielectric is independent of the electric field (usually the case for electric fields that are not too strong) and that we use the appropriate value of K for each point on the Gaussian surface.

Figure 24.22 Gauss's law with a dielectric. This figure shows a close-up of the left-hand capacitor plate in Fig. 24.15b. The Gaussian surface is a rectangular box that lies half in the conductor and half in the dielectric.



EXAMPLE 24.12 A spherical capacitor with dielectric

Use Gauss's law to find the capacitance of the spherical capacitor of Example 24.3 (Section 24.1) if the volume between the shells is filled with an insulating oil with dielectric constant K .

IDENTIFY and SET UP The spherical symmetry of the problem is not changed by the presence of the dielectric, so as in Example 24.3, we use a concentric spherical Gaussian surface of radius r between the shells. Since a dielectric is present, we use Gauss's law in the form of Eq. (24.23).

EXECUTE From Eq. (24.23),

$$\oint \vec{K}\vec{E} \cdot d\vec{A} = \oint KE dA = KE \oint dA = (KE)(4\pi r^2) = \frac{Q}{\epsilon_0}$$

$$E = \frac{Q}{4\pi K\epsilon_0 r^2} = \frac{Q}{4\pi\epsilon r^2}$$

where $\epsilon = K\epsilon_0$. Compared to the case in which there is vacuum between the shells, the electric field is reduced by a factor of $1/K$. The

potential difference V_{ab} between the shells is reduced by the same factor, and so the capacitance $C = Q/V_{ab}$ is *increased* by a factor of K , just as for a parallel-plate capacitor when a dielectric is inserted. Using the result of Example 24.3, we find that the capacitance with the dielectric is

$$C = \frac{4\pi K\epsilon_0 r_a r_b}{r_b - r_a} = \frac{4\pi\epsilon r_a r_b}{r_b - r_a}$$

EVALUATE If the dielectric fills the volume between the two conductors, the capacitance is just K times the value with no dielectric. The result is more complicated if the dielectric only partially fills this volume.

KEYCONCEPT Gauss's law can be written in a modified way if a dielectric is present. If the volume of space being studied is filled with a dielectric that has dielectric constant K , then the flux of the *product* of K and the electric field through a closed surface is proportional to the amount of *free* charge enclosed by the surface.

TEST YOUR UNDERSTANDING OF SECTION 24.6 A single point charge q is embedded in a very large block of dielectric of dielectric constant K . At a point inside the dielectric a distance r from the point charge, what is the magnitude of the electric field? (i) $q/4\pi\epsilon_0 r^2$; (ii) $Kq/4\pi\epsilon_0 r^2$; (iii) $q/4\pi K\epsilon_0 r^2$; (iv) none of these.

ANSWER

(iii) Equation (24.23) shows that this situation is the same as an isolated point charge in vacuum but with $\vec{E}$ replaced by $\vec{K}\vec{E}$. Hence KE at the point of interest is equal to $q/4\pi\epsilon_0 r^2$, and so $E = q/4\pi K\epsilon_0 r^2$. As in Example 24.12, filling the space with a dielectric reduces the electric field by a factor of $1/K$.

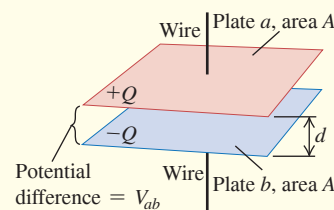
CHAPTER 24 SUMMARY

Capacitors and capacitance: A capacitor is any pair of conductors separated by an insulating material. When the capacitor is charged, there are charges of equal magnitude Q and opposite sign on the two conductors, and the potential V_{ab} of the positively charged conductor with respect to the negatively charged conductor is proportional to Q . The capacitance C is defined as the ratio of Q to V_{ab} . The SI unit of capacitance is the farad (F): $1 \text{ F} = 1 \text{ C/V}$.

A parallel-plate capacitor consists of two parallel conducting plates, each with area A , separated by a distance d . If they are separated by vacuum, the capacitance depends on only A and d . For other geometries, the capacitance can be found by using the definition $C = Q/V_{ab}$. (See Examples 24.1–24.4.)

$$C = \frac{Q}{V_{ab}} \quad (24.1)$$

$$C = \frac{Q}{V_{ab}} = \epsilon_0 \frac{A}{d} \quad (24.2)$$



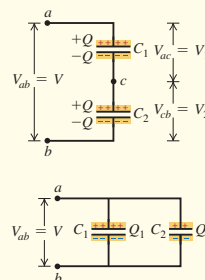
Capacitors in series and parallel: When capacitors with capacitances $C_1, C_2, C_3, \dots$ are connected in series, the reciprocal of the equivalent capacitance C_{eq} equals the sum of the reciprocals of the individual capacitances. When capacitors are connected in parallel, the equivalent capacitance C_{eq} equals the sum of the individual capacitances. (See Examples 24.5 and 24.6.)

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots \quad (24.5)$$

(capacitors in series)

$$C_{eq} = C_1 + C_2 + C_3 + \dots \quad (24.7)$$

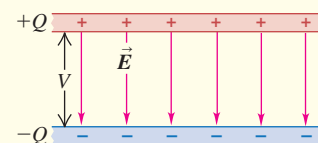
(capacitors in parallel)



Energy in a capacitor: The energy U required to charge a capacitor C to a potential difference V and a charge Q is equal to the energy stored in the capacitor. This energy can be thought of as residing in the electric field between the conductors; the energy density u (energy per unit volume) is proportional to the square of the electric-field magnitude. (See Examples 24.7–24.9.)

$$U = \frac{Q^2}{2C} = \frac{1}{2}CV^2 = \frac{1}{2}QV \quad (24.9)$$

$$u = \frac{1}{2}\epsilon_0 E^2 \quad (24.11)$$



Dielectrics: When the space between the conductors is filled with a dielectric material, the capacitance increases by a factor K , the dielectric constant of the material. The quantity $\epsilon = K\epsilon_0$ is the permittivity of the dielectric. For a fixed amount of charge on the capacitor plates, induced charges on the surface of the dielectric decrease the electric field and potential difference between the plates by the same factor K . The surface charge results from polarization, a microscopic rearrangement of charge in the dielectric. (See Example 24.10.)

Under sufficiently strong electric fields, dielectrics become conductors, a situation called dielectric breakdown. The maximum field that a material can withstand without breakdown is called its dielectric strength.

In a dielectric, the expression for the energy density is the same as in vacuum but with ϵ_0 replaced by $\epsilon = K\epsilon_0$. (See Example 24.11.)

Gauss's law in a dielectric has almost the same form as in vacuum, with two key differences: $\vec{E}$ is replaced by $K\vec{E}$ and Q_{encl} is replaced by $Q_{\text{encl-free}}$, which includes only the free charge (not bound charge) enclosed by the Gaussian surface. (See Example 24.12.)

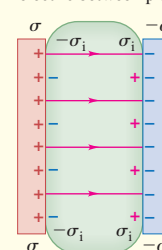
$$C = KC_0 = K\epsilon_0 \frac{A}{d} = \epsilon \frac{A}{d} \quad (24.19)$$

(parallel-plate capacitor filled with dielectric)

$$u = \frac{1}{2}K\epsilon_0 E^2 = \frac{1}{2}\epsilon E^2 \quad (24.20)$$

$$\oint K\vec{E} \cdot d\vec{A} = \frac{Q_{\text{encl-free}}}{\epsilon_0} \quad (24.23)$$

Dielectric between plates



Chapter 24 Media Assets



GUIDED PRACTICE

For assigned homework and other learning materials, go to Mastering Physics.

KEY EXAMPLE VARIATION PROBLEMS

Be sure to review **EXAMPLES 24.2, 24.3, and 24.4** (Section 24.1) before attempting these problems.

VP24.4.1 The plates of a parallel-plate capacitor in vacuum are 3.70 mm apart and 2.75 m² in area. When you apply a certain potential difference across the capacitor, the surface charge density on the positive plate is 1.40×10^{-5} C/m². Calculate (a) the capacitance of the capacitor and (b) the potential difference.

VP24.4.2 One model of a parallel-plate capacitor in vacuum is advertised as having a capacitance of 8.30 nF. When you disassemble one such capacitor, you find that its plates are 4.50 m² in area. (a) What is the spacing between this capacitor's plates? (b) When the potential difference between the plates is 3.00 kV, what are the charge on the capacitor and the magnitude of the electric field between the plates?

VP24.4.3 A spherical capacitor has vacuum between its conducting shells and a capacitance of 125 pF. The outer shell has inner radius 9.00 cm. (a) What is the outer radius of the inner shell? If the potential of the inner shell is 355 V higher than the potential of the outer shell, what are (b) the capacitor charge and (c) the surface charge densities on the inner and outer shells?

VP24.4.4 A cylindrical capacitor has a capacitance per length of 69.0 pF/m. (a) What is the ratio of the inner radius of the outer cylinder to the outer radius of the inner cylinder? (b) If the outer conductor carries charge -8.62 nC per meter of length, what is the potential difference between the two conductors? Which conductor is at higher potential?

Be sure to review **EXAMPLES 24.5 and 24.6** (Section 24.2) and **EXAMPLES 24.7, 24.8, and 24.9** (Section 24.3) before attempting these problems.

VP24.9.1 You have three capacitors that you wish to arrange in a circuit: $C_1 = 1.00$ μ F, $C_2 = 2.50$ μ F, and $C_3 = 5.00$ μ F. Find their equivalent capacitance if (a) they are connected in series; (b) they are connected in parallel; (c) C_1 and C_2 are connected in parallel, and that combination is connected to C_3 in series; (d) C_1 and C_2 are connected in series, and that combination is connected to C_3 in parallel.

VP24.9.2 You have two capacitors that you wish to use in an energy-storage device: $C_1 = 1.50$ μ F and $C_2 = 6.00$ μ F. (a) How much energy is stored in each capacitor if each has charge 4.50×10^{-4} C? Which has greater stored energy? (b) How much energy is stored in each capacitor if each has potential difference 125 V? Which has greater stored energy?

VP24.9.3 You connect two capacitors, $C_1 = 1.50$ μ F and $C_2 = 4.00$ μ F. (a) How much energy is stored in the combination if the potential difference across the combination is 145 V and the capacitors are connected in series? In parallel? In which case is the stored energy greater? (b) How much energy is stored in the combination if the charge on the combination is 4.00×10^{-4} C and the capacitors are connected in series? In parallel? In which case is the stored energy greater?

VP24.9.4 The plates of a parallel-plate capacitor in vacuum are 1.40 mm apart and 2.45 m² in area. The charge on the capacitor is 1.25×10^{-5} C. Calculate (a) the surface charge density on the positive plate, (b) the electric field magnitude between the plates, (c) the electric energy density between the plates, and (d) the total energy stored in the field between the plates.

Be sure to review **EXAMPLES 24.10 and 24.11** (Section 24.4) before attempting these problems.

VP24.11.1 The two plates of a parallel-plate capacitor each have area 0.460 m^2 , are 3.00 mm apart, and initially have vacuum between them. A power supply is attached to the capacitor, charging it to 4.00 kV , and is then disconnected. A dielectric sheet is then inserted that fills the space between the plates. The potential difference between the plates decreases to 2.50 kV , and the charge on each plate remains constant. Find (a) the original capacitance, (b) the charge on the capacitor plates, (c) the dielectric constant of the dielectric, (d) the new capacitance after you insert the dielectric, and (e) the magnitude of the induced charge on the faces of the dielectric sheet.

VP24.11.2 The two plates of a parallel-plate capacitor each have area 1.20 m^2 , are 4.50 mm apart, and initially have vacuum between them. You connect a power supply to the capacitor, charging it to 3.50 kV . Keeping the power supply connected, you insert a sheet of dielectric with dielectric constant 2.50 that fills the space between the plates.

Find (a) the original capacitance, (b) the original charge on the capacitor plates, (c) the new capacitance once you insert the dielectric, (d) the new charge on the capacitor plates, and (e) the magnitude of the induced charge on the faces of the dielectric sheet.

VP24.11.3 With vacuum between its plates, a parallel-plate capacitor has capacitance $4.50 \mu\text{F}$. You attach a power supply to the capacitor, charging it to 2.80 kV , and then disconnect it. You then insert a dielectric sheet that fills the space between the plates. The potential difference between the plates decreases to 1.20 kV , and the charge on each plate remains constant. Find the energy stored in the capacitor (a) before you insert the sheet and (b) after you insert the sheet.

VP24.11.4 With vacuum between its plates, a parallel-plate capacitor has capacitance $5.60 \mu\text{F}$. You attach a power supply to the capacitor, charging it to 1.50 kV . Keeping the power supply connected, you insert a dielectric sheet with dielectric constant 2.85 that fills the space between the plates. Find the energy stored in the capacitor (a) before the sheet is inserted and (b) after the sheet is inserted.

BRIDGING PROBLEM Electric-Field Energy and Capacitance of a Conducting Sphere

A solid conducting sphere of radius R carries a charge Q . Calculate the electric-field energy density at a point a distance r from the center of the sphere for (a) $r < R$ and (b) $r > R$. (c) Calculate the total electric-field energy associated with the charged sphere. (d) How much work is required to assemble the charge Q on the sphere? (e) Use the result of part (c) to find the capacitance of the sphere. (You can think of the second conductor as a hollow conducting shell of infinite radius.)

SOLUTION GUIDE

IDENTIFY and SET UP

1. You know the electric field for this situation at all values of r from Example 22.5 (Section 22.4). You'll use this to find the electric-field energy density u and total electric-field energy U . You can then find the capacitance from the relationship $U = Q^2/2C$.
2. To find U , consider a spherical shell of radius r and thickness dr that has volume $dV = 4\pi r^2 dr$. (It will help to make a drawing of such a shell concentric with the conducting sphere.) The energy stored in this volume is $u dV$, and the total energy is the integral of $u dV$ from $r = 0$ to $r \rightarrow \infty$. Set up this integral.

EXECUTE

3. Find u for $r < R$ and for $r > R$. (*Hint:* What is the field inside a solid conductor?)
4. Substitute your results from step 3 into the expression from step 2. Then calculate the integral to find the total electric-field energy U .
5. Use your understanding of the energy stored in a charge distribution to find the work required to assemble the charge Q .
6. Find the capacitance of the sphere.

EVALUATE

7. Where is the electric-field energy density greatest? Where is it least?
8. How would the results be affected if the solid sphere were replaced by a hollow conducting sphere of the same radius R ?
9. You can find the potential difference between the sphere and infinity from $C = Q/V$. Does this agree with the result of Example 23.8 (Section 23.3)?

PROBLEMS

•, ••, •••: Difficulty levels. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **DATA**: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. **BIO**: Biosciences problems.

DISCUSSION QUESTIONS

Q24.1 Equation (24.2) shows that the capacitance of a parallel-plate capacitor becomes larger as the plate separation d decreases. However, there is a practical limit to how small d can be made, which places limits on how large C can be. Explain what sets the limit on d . (*Hint:* What happens to the magnitude of the electric field as $d \rightarrow 0$?)

Q24.2 Suppose several different parallel-plate capacitors are charged up by a constant-voltage source. Thinking of the actual movement and position of the charges on an atomic level, why does it make sense that the capacitances are proportional to the surface areas of the plates? Why does it make sense that the capacitances are *inversely* proportional to the distance between the plates?

Q24.3 Suppose the two plates of a capacitor have different areas. When the capacitor is charged by connecting it to a battery, do the charges on the two plates have equal magnitude, or may they be different? Explain your reasoning.

Q24.4 To store the maximum amount of energy in a parallel-plate capacitor with a given battery (voltage source), would it be better to have the plates far apart or close together?

Q24.5 In the parallel-plate capacitor of Fig. 24.2, suppose the plates are pulled apart so that the separation d is much larger than the size of the plates. (a) Is it still accurate to say that the electric field between the plates is uniform? Why or why not? (b) In the situation shown in Fig. 24.2, the potential difference between the plates is $V_{ab} = Qd/\epsilon_0 A$. If the plates are pulled apart as described above, is V_{ab} more or less than this formula would indicate? Explain your reasoning. (c) With the plates pulled apart as described above, is the capacitance more than, less than, or the same as that given by Eq. (24.2)? Explain your reasoning.

Q24.6 A parallel-plate capacitor is charged by being connected to a battery and is kept connected to the battery. The separation between the plates is then doubled. How does the electric field change? The charge on the plates? The total energy? Explain.

Q24.7 A parallel-plate capacitor is charged by being connected to a battery and is then disconnected from the battery. The separation between the plates is then doubled. How does the electric field change? The potential difference? The total energy? Explain.

Q24.8 Two parallel-plate capacitors, identical except that one has twice the plate separation of the other, are charged by the same voltage source. Which capacitor has a stronger electric field between the plates? Which capacitor has a greater charge? Which has greater energy density? Explain your reasoning.

Q24.9 The charged plates of a capacitor attract each other, so to pull the plates farther apart requires work by some external force. What becomes of the energy added by this work? Explain.

Q24.10 You have two capacitors and want to connect them across a voltage source (battery) to store the maximum amount of energy. Should they be connected in series or in parallel?

Q24.11 As shown in Table 24.1, water has a very large dielectric constant $K = 80.4$. Why do you think water is not commonly used as a dielectric in capacitors?

Q24.12 Is dielectric strength the same thing as dielectric constant? Explain any differences between the two quantities. Is there a simple relationship between dielectric strength and dielectric constant (see Table 24.2)?

Q24.13 A capacitor made of aluminum foil strips separated by Mylar film was subjected to excessive voltage, and the resulting dielectric breakdown melted holes in the Mylar. After this, the capacitance was found to be about the same as before, but the breakdown voltage was much less. Why?

Q24.14 Suppose you bring a slab of dielectric close to the gap between the plates of a charged capacitor, preparing to slide it between the plates. What force will you feel? What does this force tell you about the energy stored between the plates once the dielectric is in place, compared to before the dielectric is in place?

Q24.15 The freshness of fish can be measured by placing a fish between the plates of a capacitor and measuring the capacitance. How does this work? (*Hint:* As time passes, the fish dries out. See Table 24.1.)

Q24.16 Electrolytic capacitors use as their dielectric an extremely thin layer of nonconducting oxide between a metal plate and a conducting solution. Discuss the advantage of such a capacitor over one constructed using a solid dielectric between the metal plates.

Q24.17 In terms of the dielectric constant K , what happens to the electric flux through the Gaussian surface shown in Fig. 24.22 when the dielectric is inserted into the previously empty space between the plates? Explain.

Q24.18 A parallel-plate capacitor is connected to a power supply that maintains a fixed potential difference between the plates. (a) If a sheet of dielectric is then slid between the plates, what happens to (i) the electric field between the plates, (ii) the magnitude of charge on each plate, and (iii) the energy stored in the capacitor? (b) Now suppose that before the dielectric is inserted, the charged capacitor is disconnected from the power supply. In this case, what happens to (i) the electric field between the plates, (ii) the magnitude of charge on each plate, and (iii) the energy stored in the capacitor? Explain any differences between the two situations.

Q24.19 Liquid dielectrics that have polar molecules (such as water) always have dielectric constants that decrease with increasing temperature. Why?

Q24.20 A conductor is an extreme case of a dielectric, since if an electric field is applied to a conductor, charges are free to move within the conductor to set up “induced charges.” What is the dielectric constant of a perfect conductor? Is it $K = 0$, $K \rightarrow \infty$, or something in between? Explain your reasoning.

Q24.21 The two plates of a capacitor are given charges $\pm Q$. The capacitor is then disconnected from the charging device so that the charges on the plates can’t change, and the capacitor is immersed in a tank of oil. Does the electric field between the plates increase, decrease, or stay the same? Explain your reasoning. How can this field be measured?

EXERCISES

Section 24.1 Capacitors and Capacitance

24.1 • The plates of a parallel-plate capacitor are 2.50 mm apart, and each carries a charge of magnitude 80.0 nC. The plates are in vacuum. The electric field between the plates has a magnitude of 4.00×10^6 V/m. What is (a) the potential difference between the plates; (b) the area of each plate; (c) the capacitance?

24.2 • The plates of a parallel-plate capacitor are 3.28 mm apart, and each has an area of 9.82 cm^2 . Each plate carries a charge of magnitude 4.35×10^{-8} C. The plates are in vacuum. What is (a) the capacitance; (b) the potential difference between the plates; (c) the magnitude of the electric field between the plates?

24.3 • A parallel-plate air capacitor of capacitance 245 pF has a charge of magnitude $0.148 \mu\text{C}$ on each plate. The plates are 0.328 mm apart. (a) What is the potential difference between the plates? (b) What is the area of each plate? (c) What is the electric-field magnitude between the plates? (d) What is the surface charge density on each plate?

24.4 • A $5.00 \mu\text{F}$ parallel-plate capacitor is connected to a 12.0 V battery. After the capacitor is fully charged, the battery is disconnected without loss of any of the charge on the plates. (a) A voltmeter is connected across the two plates without discharging them. What does it read? (b) What would the voltmeter read if (i) the plate separation were doubled; (ii) the radius of each plate were doubled but their separation was unchanged?

24.5 • A $10.0 \mu\text{F}$ parallel-plate capacitor with circular plates is connected to a 12.0 V battery. (a) What is the charge on each plate? (b) How much charge would be on the plates if their separation were doubled while the capacitor remained connected to the battery? (c) How much charge would be on the plates if the capacitor were connected to the 12.0 V battery after the radius of each plate was doubled without changing their separation?

24.6 • A 5.00 pF , parallel-plate, air-filled capacitor with circular plates is to be used in a circuit in which it will be subjected to potentials of up to 1.00×10^2 V. The electric field between the plates is to be no greater than 1.00×10^4 N/C. As a budding electrical engineer for Live-Wire Electronics, your tasks are to (a) design the capacitor by finding what its physical dimensions and separation must be; (b) find the maximum charge these plates can hold.

24.7 • A parallel-plate air capacitor is to store charge of magnitude 240.0 pC on each plate when the potential difference between the plates is 42.0 V. (a) If the area of each plate is 6.80 cm^2 , what is the separation between the plates? (b) If the separation between the two plates is double the value calculated in part (a), what potential difference is required for the capacitor to store charge of magnitude 240.0 pC on each plate?

24.8 • A cylindrical capacitor consists of a solid inner conducting core with radius 0.250 cm, surrounded by an outer hollow conducting tube. The two conductors are separated by air, and the length of the cylinder is 12.0 cm. The capacitance is 36.7 pF. (a) Calculate the inner radius of the hollow tube. (b) When the capacitor is charged to 125 V, what is the charge per unit length λ on the capacitor?

24.9 • A capacitor is made from two hollow, coaxial, iron cylinders, one inside the other. The inner cylinder is negatively charged and the outer is positively charged; the magnitude of the charge on each is 10.0 pC. The inner cylinder has radius 0.50 mm, the outer one has radius 5.00 mm, and the length of each cylinder is 18.0 cm. (a) What is the capacitance? (b) What applied potential difference is necessary to produce these charges on the cylinders?

24.10 •• A cylindrical capacitor has an inner conductor of radius 2.2 mm and an outer conductor of radius 3.5 mm. The two conductors are separated by vacuum, and the entire capacitor is 2.8 m long. (a) What is the capacitance per unit length? (b) The potential of the inner conductor is 350 mV higher than that of the outer conductor. Find the charge (magnitude and sign) on both conductors.

24.11 •• A spherical capacitor contains a charge of 3.30 nC when connected to a potential difference of 220 V . If its plates are separated by vacuum and the inner radius of the outer shell is 4.00 cm , calculate: (a) the capacitance; (b) the radius of the inner sphere; (c) the electric field just outside the surface of the inner sphere.

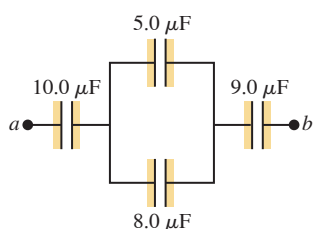
24.12 • A spherical capacitor is formed from two concentric, spherical, conducting shells separated by vacuum. The inner sphere has radius 15.0 cm and the capacitance is 116 pF . (a) What is the radius of the outer sphere? (b) If the potential difference between the two spheres is 220 V , what is the magnitude of charge on each sphere?

Section 24.2 Capacitors in Series and Parallel

24.13 •• You measure the capacitance C_1 of a capacitor by doing the following: First connect capacitors C_1 and C_2 in series to a power supply that provides a voltage V that can be varied. The capacitance of C_2 is known to be $3.00 \mu\text{F}$. Then vary the applied voltage V , and for each value of V measure the voltage V_2 across C_2 . After plotting your data as V_2 versus V , you find that the data fall close to a straight line that has slope 0.650 . What is the capacitance C_1 ?

24.14 • Figure E24.14 shows a system of four capacitors, where the potential difference across ab is 50.0 V . (a) Find the equivalent capacitance of this system between a and b . (b) How much charge is stored by this combination of capacitors? (c) How much charge is stored in each of the $10.0 \mu\text{F}$ and the $9.0 \mu\text{F}$ capacitors?

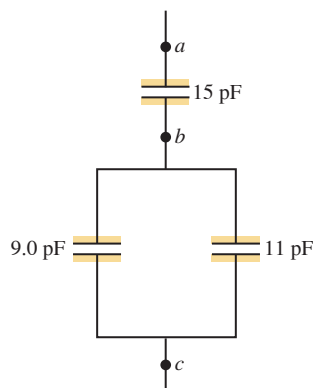
Figure E24.14



24.15 • BIO Electric Eels. Electric eels and electric fish generate large potential differences that are used to stun enemies and prey. These potentials are produced by cells that each can generate 0.10 V . We can plausibly model such cells as charged capacitors. (a) How should these cells be connected (in series or in parallel) to produce a total potential of more than 0.10 V ? (b) Using the connection in part (a), how many cells must be connected together to produce the 500 V surge of the electric eel?

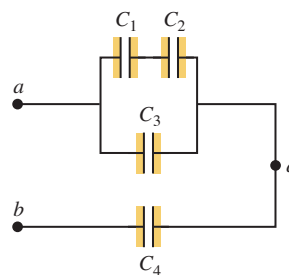
24.16 • For the system of capacitors shown in Fig. E24.16, find the equivalent capacitance (a) between b and c , and (b) between a and c .

Figure E24.16



24.17 • In Fig. E24.17, each capacitor has $C = 4.00 \mu\text{F}$ and $V_{ab} = +28.0 \text{ V}$. Calculate (a) the charge on each capacitor; (b) the potential difference across each capacitor; (c) the potential difference between points a and d .

Figure E24.17

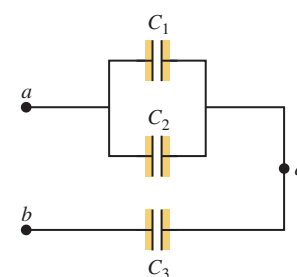


24.18 • In Fig. 24.8a, let $C_1 = 3.00 \mu\text{F}$, $C_2 = 5.00 \mu\text{F}$, and $V_{ab} = +64.0 \text{ V}$. Calculate (a) the charge on each capacitor and (b) the potential difference across each capacitor.

24.19 • In Fig. 24.9a, let $C_1 = 3.00 \mu\text{F}$, $C_2 = 5.00 \mu\text{F}$, and $V_{ab} = +52.0 \text{ V}$. Calculate (a) the charge on each capacitor and (b) the potential difference across each capacitor.

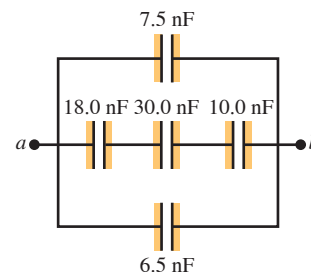
24.20 • In Fig. E24.20, $C_1 = 6.00 \mu\text{F}$, $C_2 = 3.00 \mu\text{F}$, and $C_3 = 5.00 \mu\text{F}$. The capacitor network is connected to an applied potential V_{ab} . After the charges on the capacitors have reached their final values, the charge on C_2 is $30.0 \mu\text{C}$. (a) What are the charges on capacitors C_1 and C_3 ? (b) What is the applied voltage V_{ab} ?

Figure E24.20



24.21 •• For the system of capacitors shown in Fig. E24.21, a potential difference of 25 V is maintained across ab . (a) What is the equivalent capacitance of this system between a and b ? (b) How much charge is stored by this system? (c) How much charge does the 6.5 nF capacitor store? (d) What is the potential difference across the 7.5 nF capacitor?

Figure E24.21



24.22 •• A capacitor is formed from two concentric spherical conducting shells separated by vacuum. The inner sphere has radius 12.5 cm , and the outer sphere has radius 14.8 cm . A potential difference of 120 V is applied to the capacitor. (a) What is the energy density at $r = 12.6 \text{ cm}$, just outside the inner sphere? (b) What is the energy density at $r = 14.7 \text{ cm}$, just inside the outer sphere? (c) For a parallel-plate capacitor the energy density is uniform in the region between the plates, except near the edges of the plates. Is this also true for a spherical capacitor?

Section 24.3 Energy Storage in Capacitors and Electric-Field Energy

24.23 • A $5.80 \mu\text{F}$, parallel-plate, air capacitor has a plate separation of 5.00 mm and is charged to a potential difference of 400 V . Calculate the energy density in the region between the plates, in units of J/m^3 .

24.24 • A parallel-plate air capacitor has a capacitance of 920 pF . The charge on each plate is $3.90 \mu\text{C}$. (a) What is the potential difference between the plates? (b) If the charge is kept constant, what will be the potential difference if the plate separation is doubled? (c) How much work is required to double the separation?

24.25 • An air capacitor is made from two flat parallel plates 1.50 mm apart. The magnitude of charge on each plate is $0.0180 \mu\text{C}$ when the potential difference is 200 V. (a) What is the capacitance? (b) What is the area of each plate? (c) What maximum voltage can be applied without dielectric breakdown? (Dielectric breakdown for air occurs at an electric-field strength of $3.0 \times 10^6 \text{ V/m}$.) (d) When the charge is $0.0180 \mu\text{C}$, what total energy is stored?

24.26 •• A parallel-plate vacuum capacitor has 8.38 J of energy stored in it. The separation between the plates is 2.30 mm. If the separation is decreased to 1.15 mm, what is the energy stored (a) if the capacitor is disconnected from the potential source so the charge on the plates remains constant, and (b) if the capacitor remains connected to the potential source so the potential difference between the plates remains constant?

24.27 • You have two identical capacitors and an external potential source. (a) Compare the total energy stored in the capacitors when they are connected to the applied potential in series and in parallel. (b) Compare the maximum amount of charge stored in each case. (c) Energy storage in a capacitor can be limited by the maximum electric field between the plates. What is the ratio of the electric field for the series and parallel combinations?

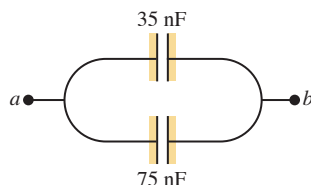
24.28 • For the capacitor network shown in Fig. E24.28, the potential difference across ab is 48 V. Find (a) the total charge stored in this network; (b) the charge on each capacitor; (c) the total energy stored in the network; (d) the energy stored in each capacitor; (e) the potential differences across each capacitor.

Figure E24.28



24.29 • For the capacitor network shown in Fig. E24.29, the potential difference across ab is 220 V. Find (a) the total charge stored in this network; (b) the charge on each capacitor; (c) the total energy stored in the network; (d) the energy stored in each capacitor; (e) the potential difference across each capacitor.

Figure E24.29



Section 24.4 Dielectrics

24.30 • Two air-filled parallel-plate capacitors with capacitances C_1 and C_2 are connected in parallel to a battery that has a voltage of 36.0 V; $C_1 = 4.00 \mu\text{F}$ and $C_2 = 6.00 \mu\text{F}$. (a) What is the total positive charge stored in the two capacitors? (b) While the capacitors remain connected to the battery, a dielectric with dielectric constant 5.00 is inserted between the plates of capacitor C_1 , completely filling the space between them. Then what is the total positive charge stored on the two capacitors? Does the insertion of the dielectric cause the total charge stored to increase or decrease?

24.31 • A $12.5 \mu\text{F}$ capacitor is connected to a power supply that keeps a constant potential difference of 24.0 V across the plates. A piece of material having a dielectric constant of 3.75 is placed between the plates, completely filling the space between them. (a) How much energy is stored in the capacitor before and after the dielectric is inserted? (b) By how much did the energy change during the insertion? Did it increase or decrease?

24.32 • A parallel-plate capacitor has capacitance $C_0 = 8.00 \text{ pF}$ when there is air between the plates. The separation between the plates is 1.50 mm. (a) What is the maximum magnitude of charge Q that can be placed on each plate if the electric field in the region between the plates is not to exceed $3.00 \times 10^4 \text{ V/m}$? (b) A dielectric with $K = 2.70$ is inserted between the plates of the capacitor, completely filling the volume between the plates. Now what is the maximum magnitude of charge on each plate if the electric field between the plates is not to exceed $3.00 \times 10^4 \text{ V/m}$?

24.33 • Two parallel plates have equal and opposite charges. When the space between the plates is evacuated, the electric field is $E = 3.20 \times 10^5 \text{ V/m}$. When the space is filled with dielectric, the electric field is $E = 2.50 \times 10^5 \text{ V/m}$. (a) What is the charge density on each surface of the dielectric? (b) What is the dielectric constant?

24.34 •• Two identical air-filled parallel-plate capacitors C_1 and C_2 , each with capacitance C , are connected in series to a battery that has voltage V . While the two capacitors remain connected to the battery, a dielectric with dielectric constant $K > 1$ is inserted between the plates of one of the capacitors, completely filling the space between them. Let U_0 be the total energy stored in the two capacitors without the dielectric and U be the total energy stored after the dielectric is inserted. In terms of K , what is the ratio U/U_0 ? Does the total stored energy increase, decrease, or stay the same after the dielectric is inserted?

24.35 •• Two identical air-filled parallel-plate capacitors C_1 and C_2 are connected in series to a battery that has voltage V . The charge on each capacitor is Q_0 . While the two capacitors remain connected to the battery, a dielectric with dielectric constant $K > 1$ is inserted between the plates of capacitor C_1 , completely filling the space between them. In terms of K and Q_0 , what is the charge on capacitor C_1 after the dielectric is inserted? Does the charge on C_1 increase, decrease, or stay the same?

24.36 •• Two air-filled parallel-plate capacitors with capacitances C_1 and C_2 are connected in series to a battery that has voltage V ; $C_1 = 3.00 \mu\text{F}$ and $C_2 = 6.00 \mu\text{F}$. The electric field between the plates of capacitor C_2 is E_{02} . While the two capacitors remain connected to the battery, a dielectric with dielectric constant $K = 4$ is inserted between the plates of capacitor C_1 , completely filling the space between them. After the dielectric is inserted in C_1 , the electric field between the plates of capacitor C_2 is E_2 . (a) What is the ratio E_2/E_{02} ? When the dielectric is inserted into C_1 , does the electric field in C_2 increase, decrease, or remain the same? (b) Repeat the calculation in part (a) for the two capacitors connected to the battery in parallel.

24.37 • The dielectric to be used in a parallel-plate capacitor has a dielectric constant of 3.60 and a dielectric strength of $1.60 \times 10^7 \text{ V/m}$. The capacitor is to have a capacitance of $1.25 \times 10^{-9} \text{ F}$ and must be able to withstand a maximum potential difference of 5500 V. What is the minimum area the plates of the capacitor may have?

24.38 •• BIO Potential in Human Cells. Some cell walls in the human body have a layer of negative charge on the inside surface and a layer of positive charge of equal magnitude on the outside surface. Suppose that the charge density on either surface is $\pm 0.50 \times 10^{-3} \text{ C/m}^2$, the cell wall is 5.0 nm thick, and the cell-wall material is air. (a) Find the magnitude of $\vec{E}$ in the wall between the two layers of charge. (b) Find the potential difference between the inside and the outside of the cell. Which is at the higher potential? (c) A typical cell in the human body has a volume of 10^{-16} m^3 . Estimate the total electric-field energy stored in the wall of a cell of this size. (Hint: Assume that the cell is spherical, and calculate the volume of the cell wall.) (d) In reality, the cell wall is made up, not of air, but of tissue with a dielectric constant of 5.4. Repeat parts (a) and (b) in this case.

24.39 • A constant potential difference of 12 V is maintained between the terminals of a $0.25 \mu\text{F}$, parallel-plate, air capacitor. (a) A sheet of Mylar is inserted between the plates of the capacitor, completely filling the space between the plates. When this is done, how much additional charge flows onto the positive plate of the capacitor (see Table 24.1)? (b) What is the total induced charge on either face of the Mylar sheet? (c) What effect does the Mylar sheet have on the electric field between the plates? Explain how you can reconcile this with the increase in charge on the plates, which acts to increase the electric field.

24.40 •• Polystyrene has dielectric constant 2.6 and dielectric strength 2.0×10^7 V/m. A piece of polystyrene is used as a dielectric in a parallel-plate capacitor, filling the volume between the plates. (a) When the electric field between the plates is 80% of the dielectric strength, what is the energy density of the stored energy? (b) When the capacitor is connected to a battery with voltage 500.0 V, the electric field between the plates is 80% of the dielectric strength. What is the area of each plate if the capacitor stores 0.200 mJ of energy under these conditions?

24.41 • When a 360 nF air capacitor ($1 \text{ nF} = 10^{-9} \text{ F}$) is connected to a power supply, the energy stored in the capacitor is $1.85 \times 10^{-5} \text{ J}$. While the capacitor is kept connected to the power supply, a slab of dielectric is inserted that completely fills the space between the plates. This increases the stored energy by $2.32 \times 10^{-5} \text{ J}$. (a) What is the potential difference between the capacitor plates? (b) What is the dielectric constant of the slab?

24.42 • A parallel-plate capacitor has capacitance $C = 12.5 \text{ pF}$ when the volume between the plates is filled with air. The plates are circular, with radius 3.00 cm. The capacitor is connected to a battery, and a charge of magnitude 25.0 pC goes onto each plate. With the capacitor still connected to the battery, a slab of dielectric is inserted between the plates, completely filling the space between the plates. After the dielectric has been inserted, the charge on each plate has magnitude 45.0 pC. (a) What is the dielectric constant K of the dielectric? (b) What is the potential difference between the plates before and after the dielectric has been inserted? (c) What is the electric field at a point midway between the plates before and after the dielectric has been inserted?

Section 24.6 Gauss's Law in Dielectrics

24.43 • A parallel-plate capacitor has the volume between its plates filled with plastic with dielectric constant K . The magnitude of the charge on each plate is Q . Each plate has area A , and the distance between the plates is d . (a) Use Gauss's law as stated in Eq. (24.23) to calculate the magnitude of the electric field in the dielectric. (b) Use the electric field determined in part (a) to calculate the potential difference between the two plates. (c) Use the result of part (b) to determine the capacitance of the capacitor. Compare your result to Eq. (24.12).

24.44 • A parallel-plate capacitor has plates with area 0.0225 m^2 separated by 1.00 mm of Teflon. (a) Calculate the charge on the plates when they are charged to a potential difference of 12.0 V. (b) Use Gauss's law [Eq. (24.23)] to calculate the electric field inside the Teflon. (c) Use Gauss's law to calculate the electric field if the voltage source is disconnected and the Teflon is removed.

PROBLEMS

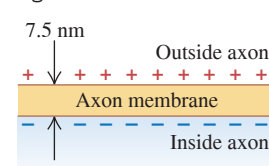
24.45 • Electronic flash units for cameras contain a capacitor for storing the energy used to produce the flash. In one such unit, the flash lasts for $\frac{1}{675} \text{ s}$ with an average light power output of $2.70 \times 10^5 \text{ W}$. (a) If the conversion of electrical energy to light is 95% efficient (the rest of the energy goes to thermal energy), how much energy must be stored in the capacitor for one flash? (b) The capacitor has a potential difference between its plates of 125 V when the stored energy equals the value calculated in part (a). What is the capacitance?

24.46 • A parallel-plate air capacitor is made by using two plates 12 cm square, spaced 3.7 mm apart. It is connected to a 12 V battery. (a) What is the capacitance? (b) What is the charge on each plate? (c) What is the electric field between the plates? (d) What is the energy stored in the capacitor? (e) If the battery is disconnected and then the plates are pulled apart to a separation of 7.4 mm, what are the answers to parts (a)–(d)?

24.47 ••• In one type of computer keyboard, each key holds a small metal plate that serves as one plate of a parallel-plate, air-filled capacitor. When the key is depressed, the plate separation decreases and the capacitance increases. Electronic circuitry detects the change in capacitance and thus detects that the key has been pressed. In one particular keyboard, the area of each metal plate is 42.0 mm^2 , and the separation between the plates is 0.700 mm before the key is depressed. (a) Calculate the capacitance before the key is depressed. (b) If the circuitry can detect a change in capacitance of 0.250 pF, how far must the key be depressed before the circuitry detects its depression?

24.48 ••• BIO Cell Membranes. Cell membranes (the walled enclosure around a cell) are typically about 7.5 nm thick. They are partially permeable to allow charged material to pass in and out, as needed. Equal but opposite charge densities build up on the inside and outside faces of such a membrane, and these charges prevent additional charges from passing through the cell wall. We can model a cell membrane as a parallel-plate capacitor, with the membrane itself containing proteins embedded in an organic material to give the membrane a dielectric constant of about 10. (See Fig. P24.48.) (a) What is the capacitance per square centimeter of such a cell wall? (b) In its normal resting state, a cell has a potential difference of 85 mV across its membrane. What is the electric field inside this membrane?

Figure P24.48

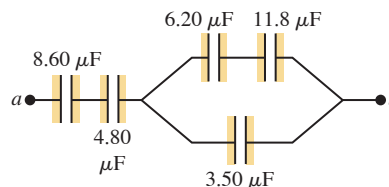


24.49 •• A $20.0 \mu\text{F}$ capacitor is charged to a potential difference of 800 V. The terminals of the charged capacitor are then connected to those of an uncharged $10.0 \mu\text{F}$ capacitor. Compute (a) the original charge of the system, (b) the final potential difference across each capacitor, (c) the final energy of the system, and (d) the decrease in energy when the capacitors are connected.

24.50 •• When lightning strikes a car, the metallic outer shell, which is insulated from the ground by its rubber tires, attains a high voltage. We can estimate how much charge is deposited by roughly modeling the car as a spherical capacitor with the outer radius taken to infinity. (a) Determine the capacitance of a sphere in terms of its radius, either by considering the potential on a sphere relative to infinity as a function of its charge, or by considering a spherical capacitor as the outer shell becomes very large. (These methods provide the same result.) (b) Estimate the radius of a sphere that corresponds to the size of a car. (c) Determine the corresponding capacitance. (d) A typical lightning strike could deliver a 100 MV potential to a car. Then what net charge would be deposited on the car?

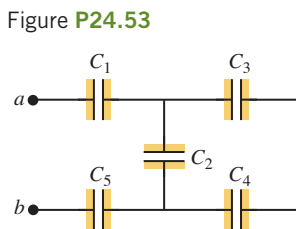
24.51 • For the capacitor network shown in Fig. P24.51, the potential difference across ab is 12.0 V. Find (a) the total energy stored in this network and (b) the energy stored in the $4.80 \mu\text{F}$ capacitor.

Figure P24.51



24.52 •• In Fig. E24.17, $C_1 = 6.00 \mu\text{F}$, $C_2 = 3.00 \mu\text{F}$, $C_3 = 4.00 \mu\text{F}$, and $C_4 = 8.00 \mu\text{F}$. The capacitor network is connected to an applied potential difference V_{ab} . After the charges on the capacitors have reached their final values, the voltage across C_3 is 40.0 V. What are (a) the voltages across C_1 and C_2 , (b) the voltage across C_4 , and (c) the voltage V_{ab} applied to the network?

24.53 • In Fig. P24.53, $C_1 = C_5 = 8.4 \mu\text{F}$ and $C_2 = C_3 = C_4 = 4.2 \mu\text{F}$. The applied potential is $V_{ab} = 220 \text{ V}$. (a) What is the equivalent capacitance of the network between points a and b ? (b) Calculate the charge on each capacitor and the potential difference across each capacitor.



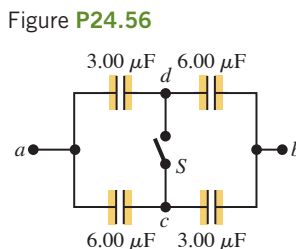
24.54 •• CP After combing your hair on a dry day, some of your hair stands up, forced away from your head by electrostatic repulsion. (a) Estimate the length L of your hair. (b) Using the average linear mass density of hair, which is $65 \mu\text{g}/\text{cm}$, estimate the mass m of one of your hairs. (c) Estimate the number N of hairs that stand after combing. (d) Assume that the comb has taken away a charge $-2Q$, and that your hair has therefore gained an amount of charge $2Q$. Assume further that half of this charge resides next to your head and the other half is distributed at the ends of the N strands that stand up. Assume the electrostatic force that lifted a hair was twice its weight. Show that this leads to

$$2mg = \frac{1}{4\pi\epsilon_0} \frac{Q^2/N}{L^2}$$

Use this equation to estimate the charge Q that resides on your head. (e) If your head were a sphere with radius R , it would have a capacitance of $4\pi\epsilon_0 R$. Estimate the radius of your head; then use your estimate to determine your head's capacitance. (f) Use your result to estimate the potential attained due to combing. (The surprising result illustrates an interesting feature of static electricity.)

24.55 •• In Fig. E24.20, $C_1 = 3.00 \mu\text{F}$ and $V_{ab} = 150 \text{ V}$. The charge on capacitor C_1 is $150 \mu\text{C}$ and the charge on C_3 is $450 \mu\text{C}$. What are the values of the capacitances of C_2 and C_3 ?

24.56 • The capacitors in Fig. P24.56 are initially uncharged and are connected, as in the diagram, with switch S open. The applied potential difference is $V_{ab} = +210 \text{ V}$. (a) What is the potential difference V_{cd} ? (b) What is the potential difference across each capacitor after switch S is closed? (c) How much charge flowed through the switch when it was closed?

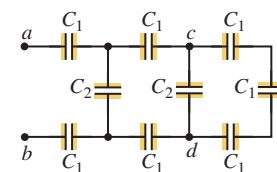


24.57 •• Three capacitors having capacitances of 8.4 , 8.4 , and $4.2 \mu\text{F}$ are connected in series across a 36 V potential difference. (a) What is the charge on the $4.2 \mu\text{F}$ capacitor? (b) What is the total energy stored in all three capacitors? (c) The capacitors are disconnected from the potential difference without allowing them to discharge. They are then reconnected in parallel with each other, with the positively charged plates connected together. What is the voltage across each capacitor in the parallel combination? (d) What is the total energy now stored in the capacitors?

24.58 • Capacitance of a Thundercloud. The charge center of a thundercloud, drifting 3.0 km above the earth's surface, contains 20 C of negative charge. Assuming the charge center has a radius of 1.0 km , and modeling the charge center and the earth's surface as parallel plates, calculate: (a) the capacitance of the system; (b) the potential difference between charge center and ground; (c) the average strength of the electric field between cloud and ground; (d) the electrical energy stored in the system.

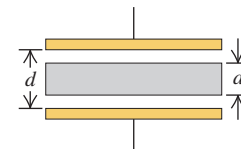
24.59 •• In Fig. P24.59, each capacitance C_1 is $6.9 \mu\text{F}$, and each capacitance C_2 is $4.6 \mu\text{F}$. (a) Compute the equivalent capacitance of the network between points a and b . (b) Compute the charge on each of the three capacitors nearest a and b when $V_{ab} = 420 \text{ V}$. (c) With 420 V across a and b , compute V_{cd} .

Figure P24.59



24.60 •• An air capacitor is made by using two flat plates, each with area A , separated by a distance d . Then a metal slab having thickness a (less than d) and the same shape and size as the plates is inserted between them, parallel to the plates and not touching either plate (Fig. P24.60).

Figure P24.60



(a) What is the capacitance of this arrangement? (b) Express the capacitance as a multiple of the capacitance C_0 when the metal slab is not present. (c) Discuss what happens to the capacitance in the limits $a \rightarrow 0$ and $a \rightarrow d$.

24.61 •• A potential difference $V_{ab} = 48.0 \text{ V}$ is applied across the capacitor network of Fig. E24.17. If $C_1 = C_2 = 4.00 \mu\text{F}$ and $C_4 = 8.00 \mu\text{F}$, what must the capacitance C_3 be if the network is to store $2.90 \times 10^{-3} \text{ J}$ of electrical energy?

24.62 • CALC The inner cylinder of a long, cylindrical capacitor has radius r_a and linear charge density $+\lambda$. It is surrounded by a coaxial cylindrical conducting shell with inner radius r_b and linear charge density $-\lambda$ (see Fig. 24.6). (a) What is the energy density in the region between the conductors at a distance r from the axis? (b) Integrate the energy density calculated in part (a) over the volume between the conductors in a length L of the capacitor to obtain the total electric-field energy per unit length. (c) Use Eq. (24.9) and the capacitance per unit length calculated in Example 24.4 (Section 24.1) to calculate U/L . Does your result agree with that obtained in part (b)?

24.63 ••• CP Dielectric elastomers are used to create voltage-dependent capacitors. These elastomers contract when subject to the stresses induced by Coulomb forces between capacitor plates. Consider a pair of parallel plates with area A separated by distance d_0 when no voltage is applied. A material with dielectric constant K and Young's modulus Y fills the space between the plates. (a) What compressive force is applied to the dielectric when the plates have opposite charges Q and $-Q$? (b) The distance between the plates is $d = d_0 - sQ^2$, where s is a squeezing coefficient. Determine s in terms of the parameters specified. (c) A capacitor of this sort has plate area 1.00 cm^2 and noncharged separation distance $d_0 = 0.400 \text{ mm}$. The plates are separated by a silicone dielectric elastomer with dielectric constant $K = 3.00$ and Young's modulus 0.0100 GPa . What is the squeezing coefficient in this case? (d) If each plate has charge of magnitude $0.700 \mu\text{C}$, what is the applied voltage? (e) What new voltage would double the amount of charge on the plates?

24.64 •• A parallel-plate capacitor is made from two plates 12.0 cm on each side and 4.50 mm apart. Half of the space between these plates contains only air, but the other half is filled with Plexiglas® of dielectric constant 3.40 (Fig. P24.64). An

Figure P24.64

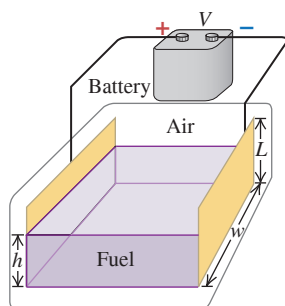


18.0 V battery is connected across the plates. (a) What is the capacitance of this combination? (Hint: Can you think of this capacitor as equivalent to two capacitors in parallel?) (b) How much energy is stored in the capacitor? (c) If we remove the Plexiglas but change nothing else, how much energy will be stored in the capacitor?

24.65 •• CP CALC A metallic circular plate with radius r is fixed to a tabletop. An identical circular plate supported from above by a cable is fixed in place a distance d above the first plate. Assume that d is much smaller than r . The two plates are attached by wires to a battery that supplies voltage V . (a) What is the tension in the cable? Neglect the weight of the plate. (b) The upper plate is slowly raised to a new height $2d$. Determine the work done by the cable by integrating $\int_d^{2d} F(z) dz$, where $F(z)$ is the cable tension when the plates are separated by a distance z . (c) Compute the energy stored in the electric field before the top plate was raised. (d) Compute the energy stored in the electric field after the top plate was raised. (e) Is the work done by the cable equal to the change in the stored electrical energy? If not, why not?

24.66 •• A fuel gauge uses a capacitor to determine the height of the fuel in a tank. The effective dielectric constant K_{eff} changes from a value of 1 when the tank is empty to a value of K , the dielectric constant of the fuel, when the tank is full. The appropriate electronic circuitry can determine the effective dielectric constant of the combined air and fuel between the capacitor plates. Each of the two rectangular plates has a width w and a length L (Fig. P24.66). The height of the fuel between the plates is h .

Figure P24.66



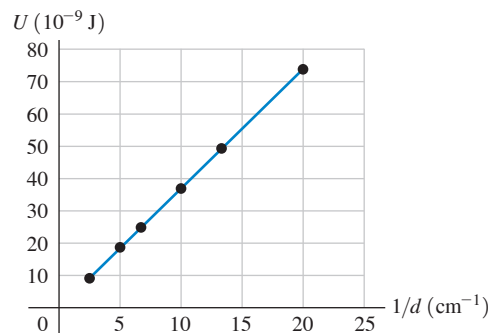
You can ignore any fringing effects. (a) Derive an expression for K_{eff} as a function of h . (b) What is the effective dielectric constant for a tank $\frac{1}{4}$ full, $\frac{1}{2}$ full, and $\frac{3}{4}$ full if the fuel is gasoline ($K = 1.95$)? (c) Repeat part (b) for methanol ($K = 33.0$). (d) For which fuel is this fuel gauge more practical?

24.67 •• DATA Your electronics company has several identical capacitors with capacitance C_1 and several others with capacitance C_2 . You must determine the values of C_1 and C_2 but don't have access to C_1 and C_2 individually. Instead, you have a network with C_1 and C_2 connected in series and a network with C_1 and C_2 connected in parallel. You have a 200.0 V battery and instrumentation that measures the total energy supplied by the battery when it is connected to the network. When the parallel combination is connected to the battery, 0.180 J of energy is stored in the network. When the series combination is connected, 0.0400 J of energy is stored. You are told that C_1 is greater than C_2 . (a) Calculate C_1 and C_2 . (b) For the series combination, does C_1 or C_2 store more charge, or are the values equal? Does C_1 or C_2 store more energy, or are the values equal? (c) Repeat part (b) for the parallel combination.

24.68 •• DATA You are designing capacitors for various applications. For one application, you want the maximum possible stored energy. For another, you want the maximum stored charge. For a third application, you want the capacitor to withstand a large applied voltage without dielectric breakdown. You start with an air-filled parallel-plate capacitor that has $C_0 = 6.00$ pF and a plate separation of 2.50 mm. You then consider the use of each of the dielectric materials listed in Table 24.2. In each application, the dielectric will fill the volume between the plates, and the electric field between the plates will be 50% of the dielectric strength given in the table. (a) For each of the five materials given in the table, calculate the energy stored in the capacitor. Which dielectric allows the maximum stored energy? (b) For each material, what is the charge Q stored on each plate of the capacitor? (c) For each material, what is the voltage applied across the capacitor? (d) Is one dielectric material in the table your best choice for all three applications?

24.69 •• DATA You are conducting experiments with an air-filled parallel-plate capacitor. You connect the capacitor to a battery with voltage 24.0 V. Initially the separation d between the plates is 0.0500 cm. In one experiment, you leave the battery connected to the capacitor, increase the separation between the plates, and measure the energy stored in the capacitor for each value of d . In a second experiment, you make the same measurements but disconnect the battery before you change the plate separation. One set of your data is given in Fig. P24.69, where you have plotted the stored energy U versus $1/d$. (a) For which experiment does this data set apply: the first (battery remains connected) or the second (battery disconnected before d is changed)? Explain. (b) Use the data plotted in Fig. P24.69 to calculate the area A of each plate. (c) For which case, the battery connected or the battery disconnected, is there more energy stored in the capacitor when $d = 0.400$ cm? Explain.

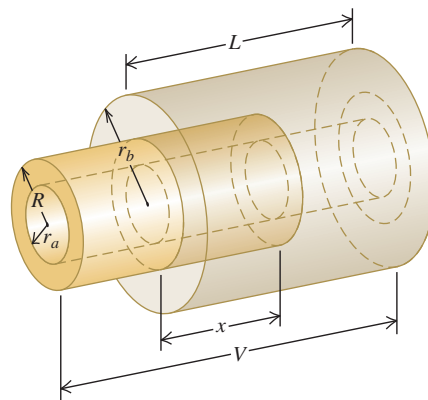
Figure P24.69



CHALLENGE PROBLEMS

24.70 ••• A thin-walled, hollow, conducting cylinder with radius r_b is concentric with a solid conducting cylinder with radius $r_a < r_b$. Each has length L . The two cylinders are attached by conducting wires to the terminals of a battery that supplies potential V . A solid cylindrical shell, with inner radius r_a and outer radius $R < r_b$, made of a material with dielectric constant K , slides between the conducting cylinders, as shown in Fig. P24.70. By changing the insertion distance x , we can alter the capacitance seen by the battery and therefore alter the amount of charge stored in this device. (a) Determine the capacitance as a function of x . (b) If $L = 10.0$ cm, $r_a = 1.00$ cm, $r_b = 4.00$ cm, $R = 3.00$ cm, and $K = 3.21$, what is the capacitance when $x = 0$? (c) What is the capacitance when $x = L$? (d) What value of x results in 6.00 nC of charge on the positively charged cylinder plate when $V = 1.00$ kV?

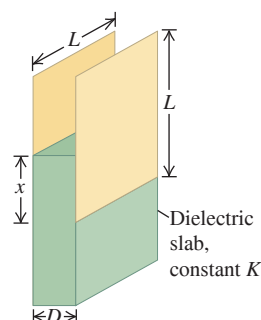
Figure P24.70



24.71 ••• **CALC** Two conducting plates with area A are separated by distance d . Between the plates is a material with a dielectric constant that varies linearly from a value of unity next to one plate to K next to the other plate. (a) What is the capacitance of this device? [Hint: We can envision this as a continuum of capacitors with differential plate separation connected in series. The reciprocal of the capacitance of a differential slice is then $dx/(K(x)\epsilon_0 A)$, where $K(x)$ is the dielectric constant specific to that locale.] (b) Show that this result matches the expected result when $K \rightarrow 1$.

24.72 ••• Two square conducting plates with sides of length L are separated by a distance D . A dielectric slab with constant K with dimensions $L \times L \times D$ is inserted a distance x into the space between the plates, as shown in **Fig. P24.72**. (a) Find the capacitance C of this system. (b) Suppose that the capacitor is connected to a battery that maintains a constant potential difference V between the plates. If the dielectric slab is inserted an additional distance dx into the space between the plates, show that the change in stored energy is

Figure P24.72



$$dU = + \frac{(K - 1)\epsilon_0 V^2 L}{2D} dx$$

(c) Suppose that before the slab is moved by dx , the plates are disconnected from the battery, so that the charges on the plates remain constant. Determine the magnitude of the charge on each plate, and then show that when the slab is moved dx farther into the space between the plates, the stored energy changes by an amount that is the *negative* of the expression for dU given in part (b). (d) If F is the force exerted on the slab by the charges on the plates, then dU should equal the work done *against* this force to move the slab a distance dx . Thus $dU = -F dx$. Show that applying this expression to the result of part (b) suggests that the electric force on the slab pushes it *out* of the capacitor, while the result of part (c) suggests that the force pulls the slab *into* the capacitor. (e) Figure 24.16 shows that the force in fact pulls the slab into the capacitor. Explain why the result of part (b) gives an incorrect answer for the direction of this force, and calculate the magnitude of the force. (This method does not require knowledge of the nature of the fringing field.)

MCAT-STYLE PASSAGE PROBLEMS

BIO The Electric Egg. Upon fertilization, the eggs of many species undergo a rapid change in potential difference across their outer membrane. This change affects the physiological development of the eggs. The potential difference across the membrane is called the *membrane potential*, V_m , which is the potential inside the membrane minus the potential outside it. The membrane potential arises when enzymes use the energy available in ATP to expel three sodium ions (Na^+) actively and accumulate two potassium ions (K^+) inside the membrane—making the interior less positively charged than the exterior. For a sea urchin egg, V_m is about -70 mV; that is, the potential inside is 70 mV less than that outside. The egg membrane behaves as a capacitor with a capacitance of about $1 \mu\text{F}/\text{cm}^2$. The membrane of the unfertilized egg is *selectively permeable* to K^+ ; that is, K^+ can readily pass through certain channels in the membrane, but other ions cannot. When a sea urchin egg is fertilized, Na^+ channels in the membrane open, Na^+ enters the egg, and V_m rapidly increases to $+30$ mV, where it remains for several minutes. The concentration of Na^+ is about 30 mmol/L in the egg's interior but 450 mmol/L in the surrounding seawater. The K^+ concentration is about 200 mmol/L inside but 10 mmol/L outside. A useful constant that connects electrical and chemical units is the *Faraday number*, which has a value of approximately 10^5 C/mol; that is, Avogadro's number (a mole) of monovalent ions, such as Na^+ or K^+ , carries a charge of 10^5 C.

24.73 How many moles of Na^+ must move per unit area of membrane to change V_m from -70 mV to $+30$ mV, if we assume that the membrane behaves purely as a capacitor? (a) 10^{-4} mol/ cm^2 ; (b) 10^{-9} mol/ cm^2 ; (c) 10^{-12} mol/ cm^2 ; (d) 10^{-14} mol/ cm^2 .

24.74 Suppose that the egg has a diameter of $200 \mu\text{m}$. What fractional change in the internal Na^+ concentration results from the fertilization-induced change in V_m ? Assume that Na^+ ions are distributed throughout the cell volume. The concentration increases by (a) 1 part in 10^4 ; (b) 1 part in 10^5 ; (c) 1 part in 10^6 ; (d) 1 part in 10^7 .

24.75 Suppose that the change in V_m was caused by the entry of Ca^{2+} instead of Na^+ . How many Ca^{2+} ions would have to enter the cell per unit membrane to produce the change? (a) Half as many as for Na^+ ; (b) the same as for Na^+ ; (c) twice as many as for Na^+ ; (d) cannot say without knowing the inside and outside concentrations of Ca^{2+} .

24.76 What is the minimum amount of work that must be done by the cell to restore V_m to -70 mV? (a) 3 mJ; (b) 3 μJ ; (c) 3 nJ; (d) 3 pJ.

ANSWERS

Chapter Opening Question ?

(iv) Equation (24.9) shows that the energy stored in a capacitor with capacitance C and charge Q is $U = Q^2/2C$. If Q is doubled, the stored energy increases by a factor of $2^2 = 4$. Note that if the value of Q is too great, the electric-field magnitude inside the capacitor will exceed the dielectric strength of the material between the plates and dielectric breakdown will occur (see Section 24.4). This puts a practical limit on the amount of energy that can be stored.

Key Example VARIATION Problems

VP24.4.1 (a) 6.58×10^{-9} F (b) 5.85 kV

VP24.4.2 (a) 4.80 mm (b) $Q = 2.49 \times 10^{-5}$ C, $E = 6.25 \times 10^5$ V/m

VP24.4.3 (a) 8.33 cm (b) 4.44×10^{-8} C

(c) $\sigma_{\text{inner}} = +5.09 \times 10^{-7}$ C/ m^2 , $\sigma_{\text{outer}} = -4.36 \times 10^{-7}$ C/ m^2

VP24.4.4 (a) 2.24 (b) 125 V; inner conductor

VP24.9.1 (a) 0.625 μF (b) 8.50 μF (c) 2.06 μF (d) 5.71 μF

VP24.9.2 (a) $U_1 = 0.0675$ J, $U_2 = 0.0169$ J; C_1 (b) $U_1 = 0.0117$ J, $U_2 = 0.0469$ J; C_2

VP24.9.3 (a) series 0.0115 J, parallel 0.0578 J; parallel (c) series 0.0733 J, parallel 0.0145 J; series

VP24.9.4 (a) 5.10×10^{-6} C/ m^2 (b) 5.76×10^5 V/m (c) 1.47 J/ m^3 (d) 5.04×10^{-3} J

VP24.11.1 (a) 1.36 nF (b) 5.43×10^{-6} C (c) 1.60 (d) 2.17 nF (e) 2.04×10^{-6} C

VP24.11.2 (a) 2.36 nF (b) 8.26×10^{-6} C (c) 5.90 nF (d) 2.07×10^{-5} C (e) 1.24×10^{-5} C

VP24.11.3 (a) 17.6 J (b) 7.56 J

VP24.11.4 (a) 6.30 J (b) 18.0 J

Bridging Problem

(a) 0

(b) $Q^2/32\pi^2\epsilon_0 r^4$

(c) $Q^2/8\pi\epsilon_0 R$

(d) $Q^2/8\pi\epsilon_0 R$

(e) $C = 4\pi\epsilon_0 R$